k-ALM: Toolkit

Liquidity Stress Testing (LST) User Guide

Version 2.2 22 February 2024 kalm@katalysys.com

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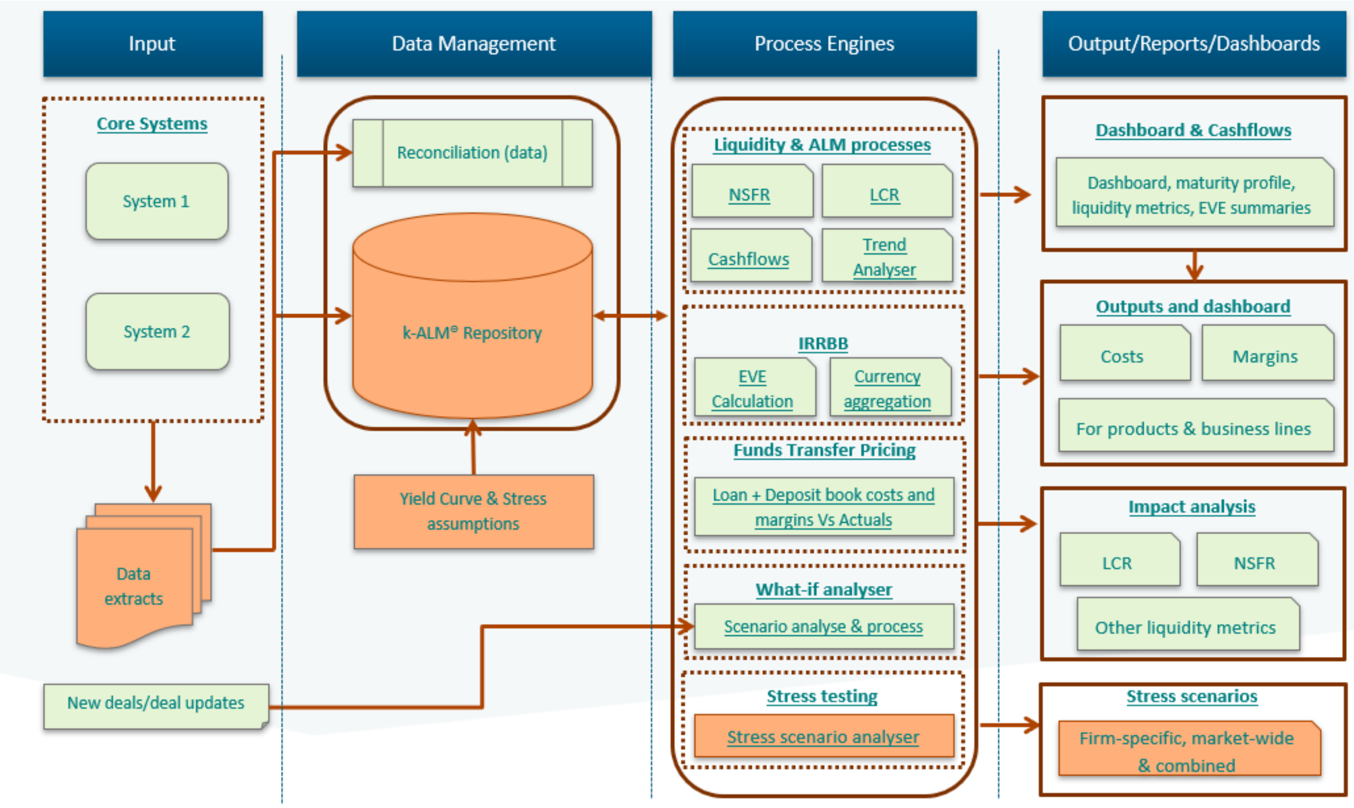
TECHNICAL COVERAGE

1 Introduction

k-ALM is a highly customisable software framework and tool-kit developed by Katalysys. This can be used for stress testing, evaluating interest rate risk on the banking book and funds transfer pricing.

The data from the bank's various core banking systems the allocation/output files from regulatory reporting systems are extracted, which is then used to conduct the stress tests.

A high-level schematic diagram of the overall k-ALM tool-kit is provided below:



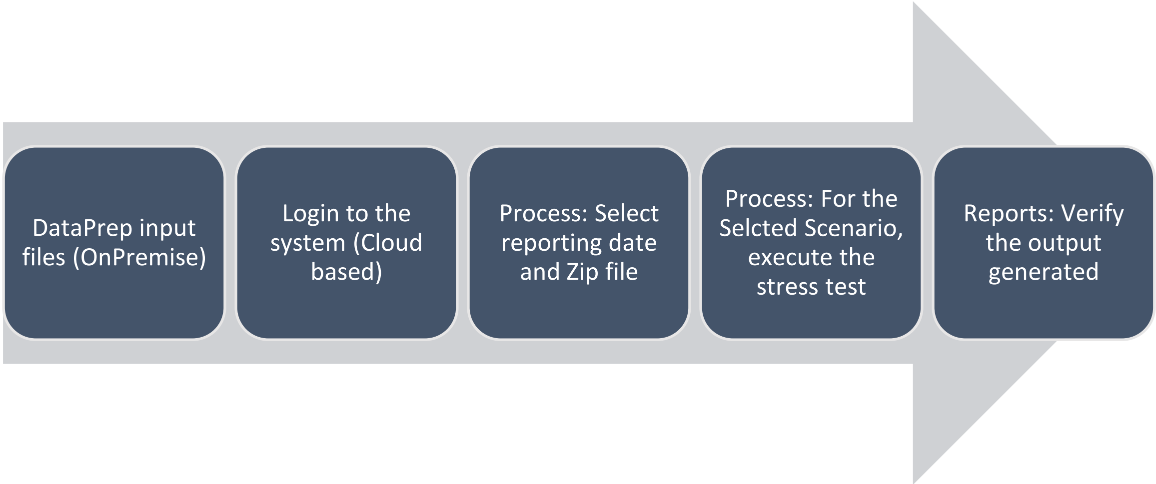
(only the elements for k-ALM Liquidity Stress Testing are highlighted in orange in the diagram above).

2 Process Flow

The process to upload the data and generate the reports/metrics is controlled via the k-ALM application. There are two parts to this

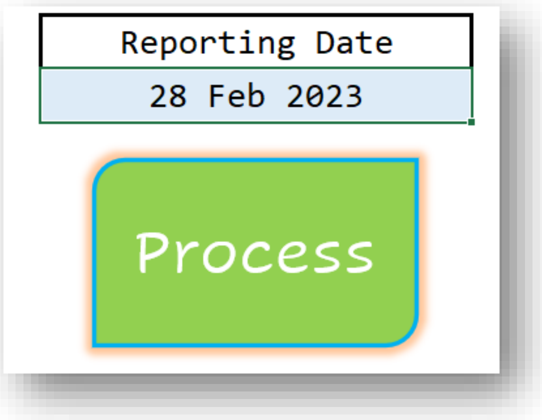
1) **ON PREMISE:** The desensitising of the data – which removes customer sensitive information (e.g. customer identified, account number, customer name)
2) **CLOUD:** Uploading and processing the data to generate the reports/metrics

The steps involved are shown below



2.1 DataPrep Macro (On premise)

1) Ensure that all the data extract(s) are in the **DATA** folder.
2) Open the DataPrep Excel Macro (**kALM_BankId_DataPrepTool_n.n[1].xlsm**)
3) Ensure direct entries and Parameters worksheets are up to date with the required data.
4) Enter the 'Reporting Date' (blue shaded cell) and Click on the 'PROCESS' button

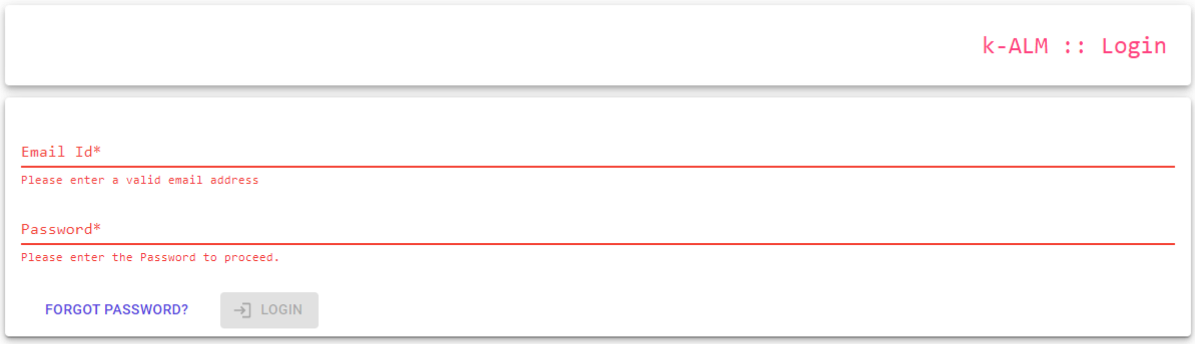


5) Once the Process is complete, a message is displayed.

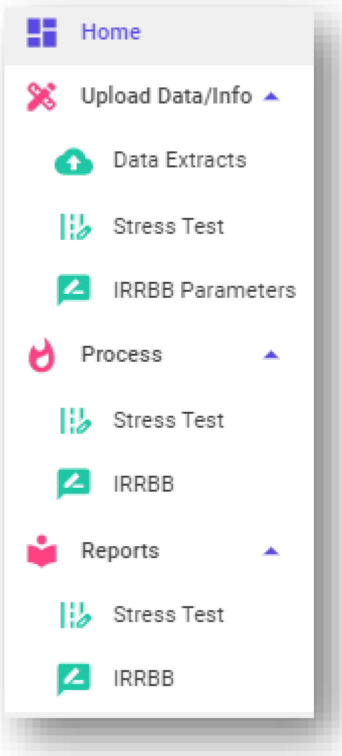
[1] Where n.n is the version number.

2.2 Executing the stress scenario(s) [Cloud based]

1) **Login** to the system [**https://bankname.k-alm.com/nnnnnnn[2]/Login**] with your user-id and password



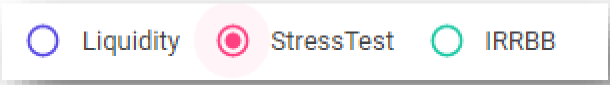
On the left side of the screen a navigation bar is provided



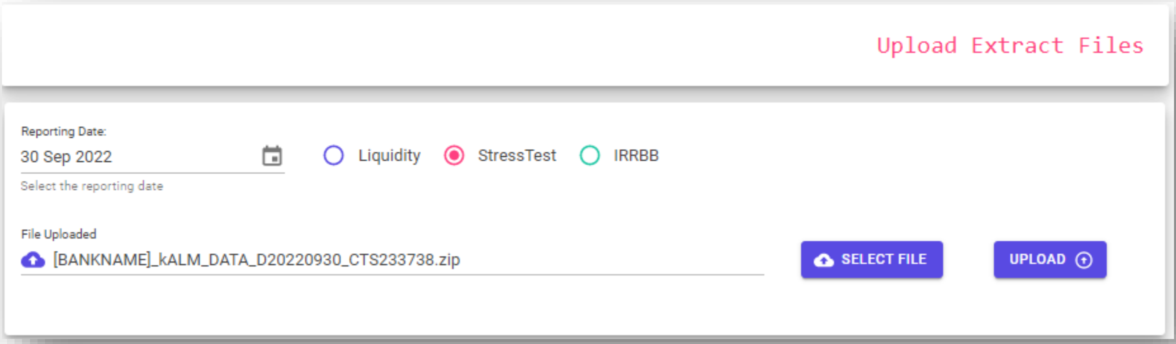
2) To begin the process, navigate to **Upload Data/Info → Data Extracts,** this is where the extract files can be uploaded for the specific Reporting Date.

[2] Each bank is provided a unique id for enhanced security

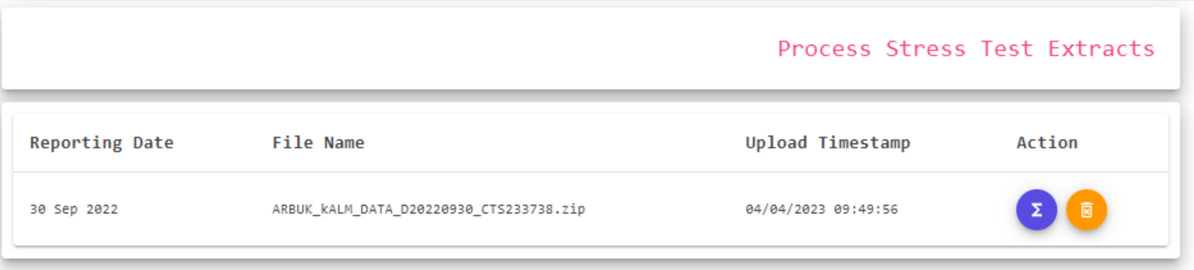
3) Ensure that **StressTest** is selected.



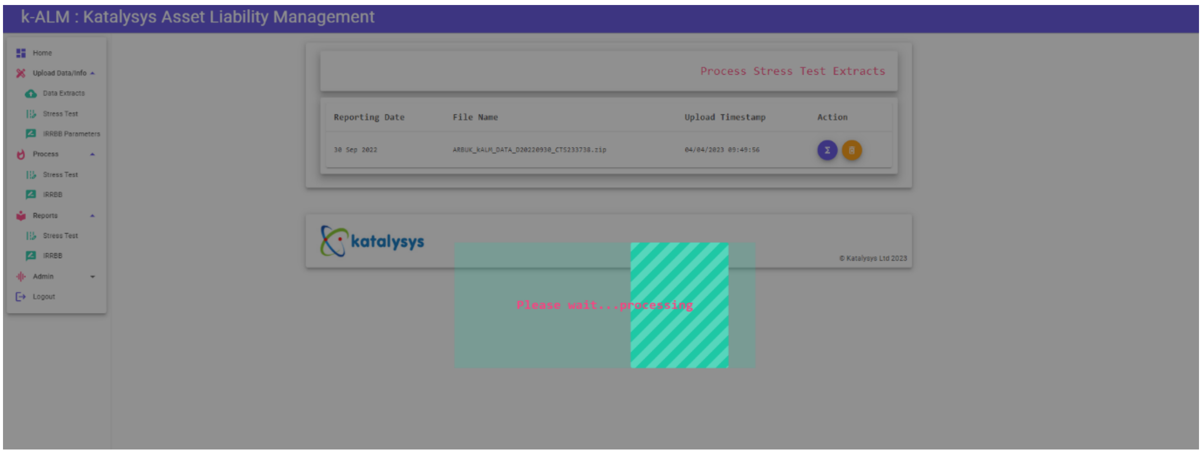
4) Select the **Reporting Date** and the corresponding Zip file (by clicking on 'SELECT FILES') to process and then click on 'UPLOAD NOW'.



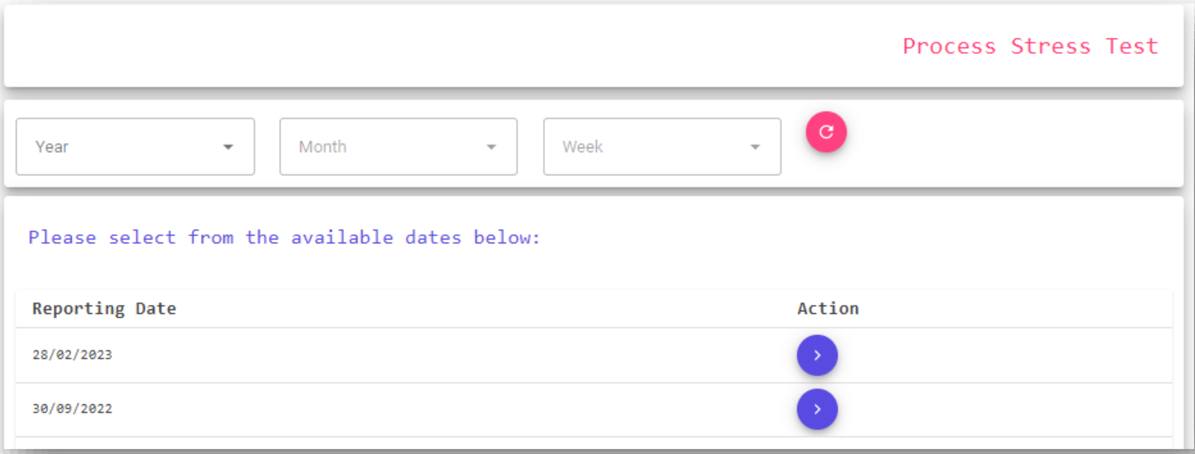
5) The file that is uploaded is now ready for Stress Test mapping. Click on the **Purple button**, found under the Action column, as shown below.



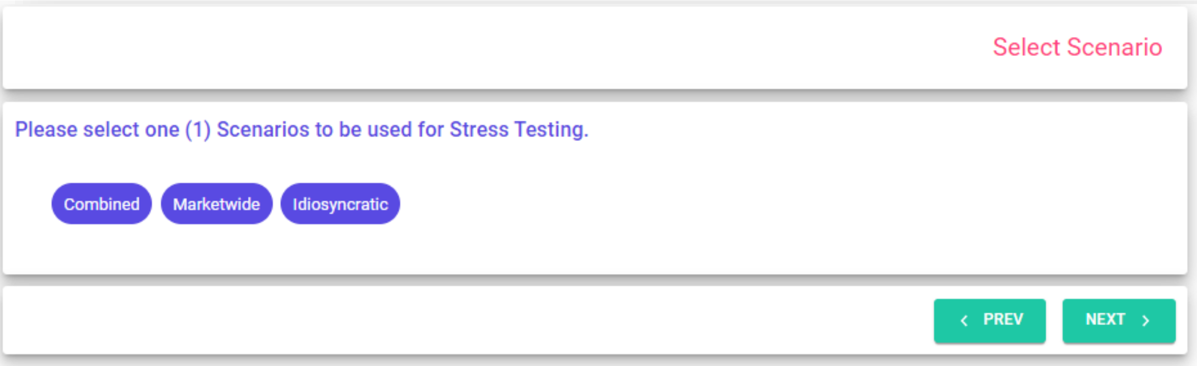
6) Once clicked, the process will execute, and the following screen will be shown.

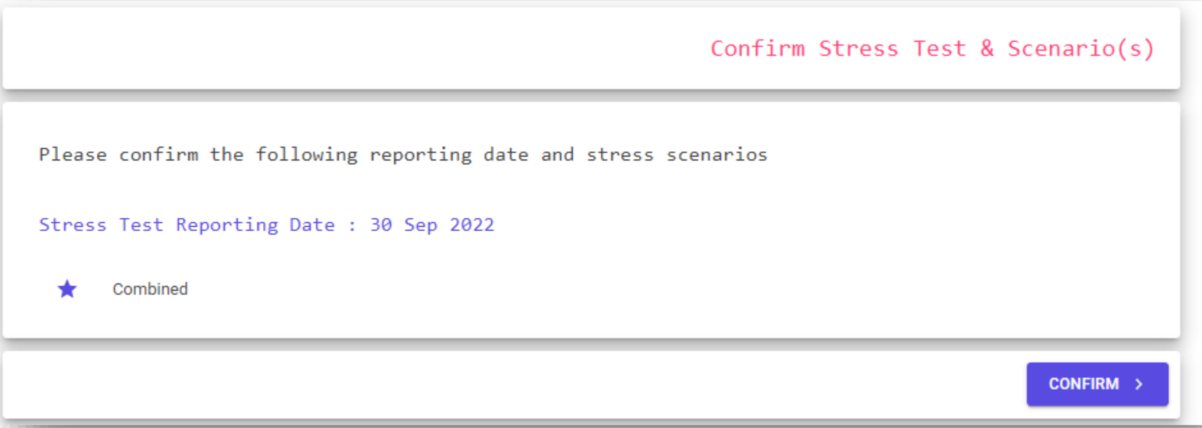


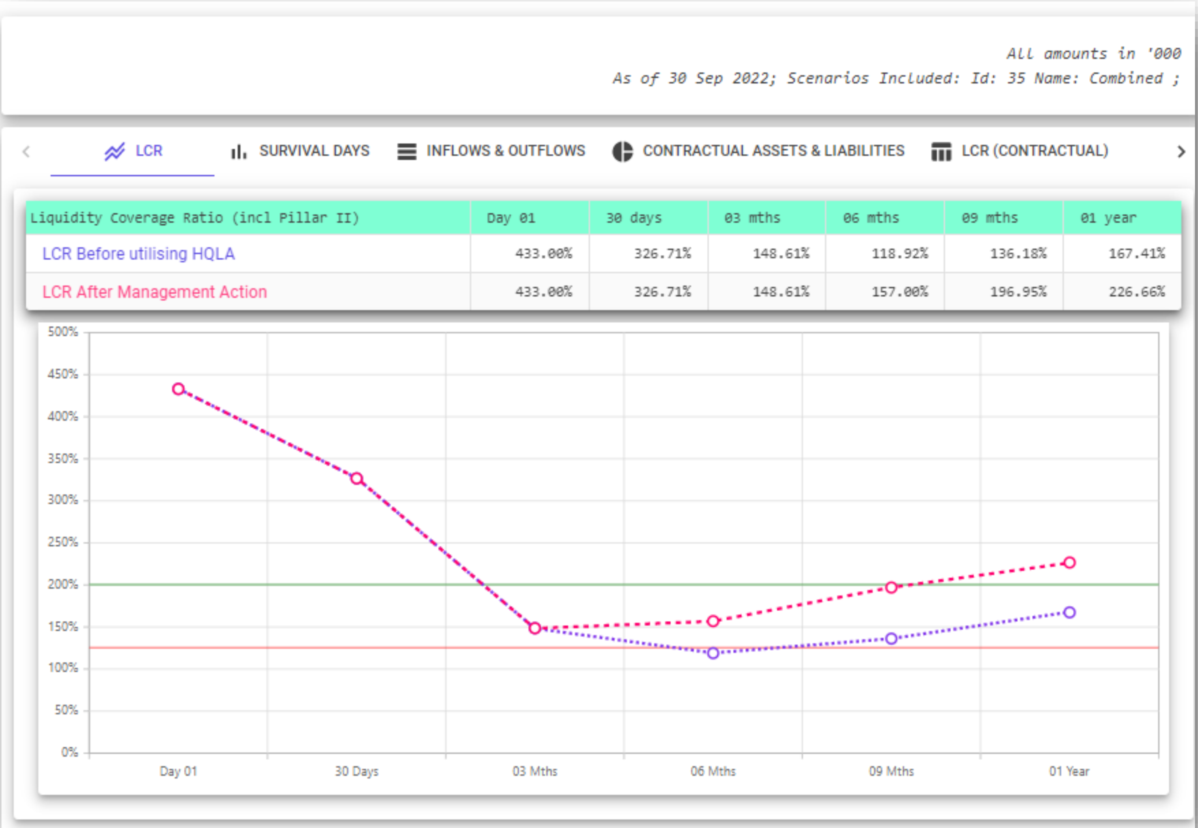
7) Once completed the screen will automatically transfer to the **Process** stage. If no initial upload is required and the user would like to process an existing upload file, Proceed to **Process → Stress Test**.
Identify the date for which the stress test has to be run and click the **Purple Arrow Button** under Action.



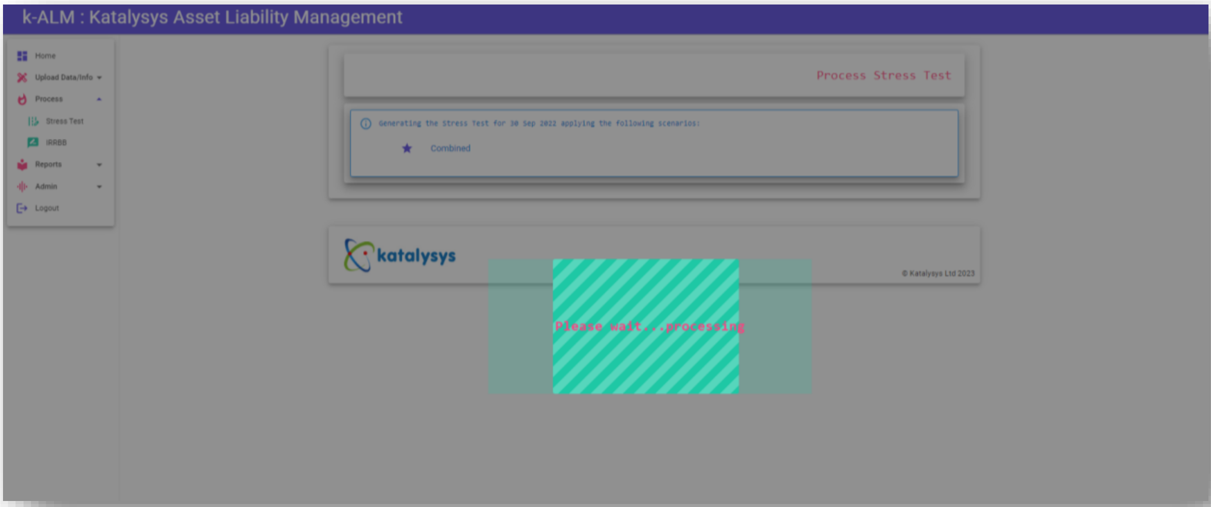
8) Select the scenario to be evaluated and click 'NEXT'





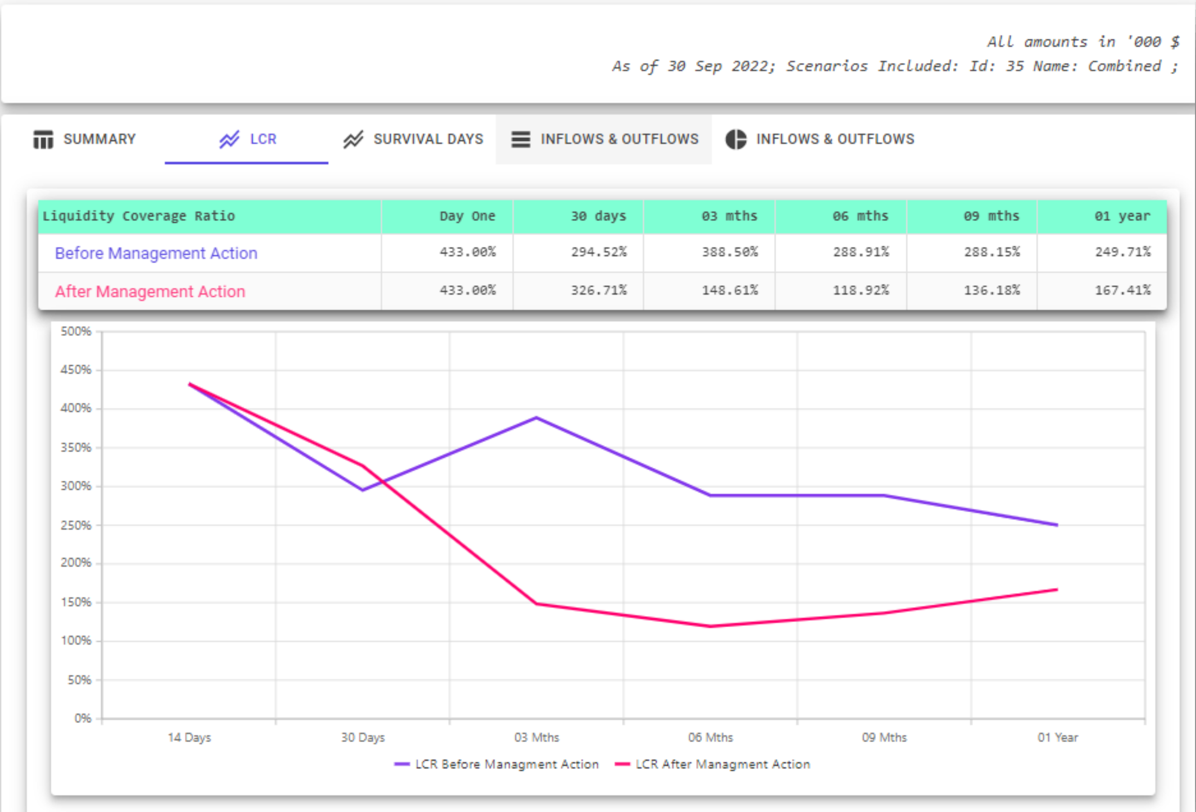


9) Confirm the selection. Once verified click 'CONFIRM', and the process will be executed Once 'CONFIRM' has been selected, the process will execute, and the following screen will be shown

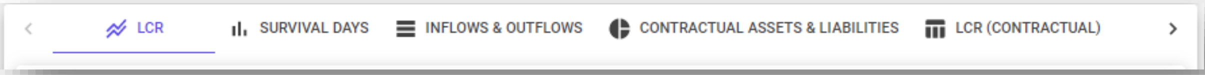


10) Upon completion of the process, the dashboard with the results are shown automatically, starting with impact of the LCR over the 1 year period.

An example is shown below:



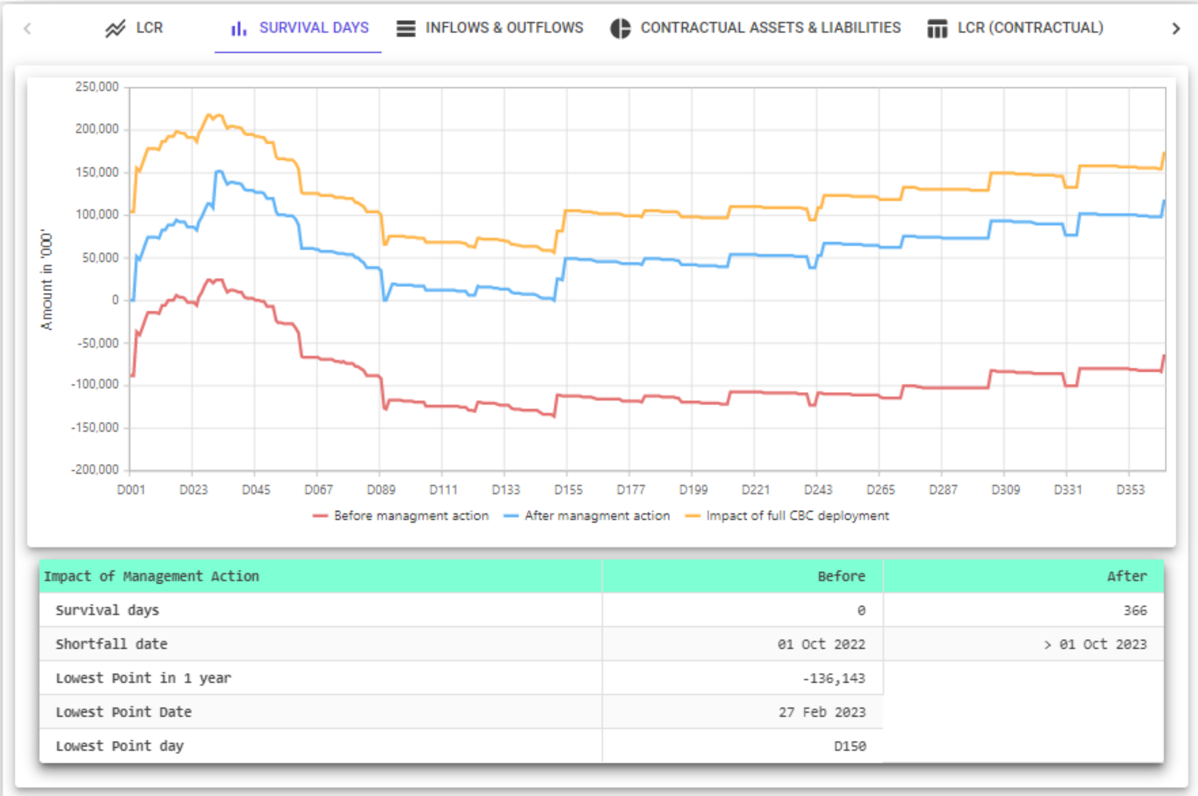
11) The user can switch between the different assessments by clicking on the tabs, located just above the initial results table, as shown below:



Use the arrows on the right and left hand side to navigate, there are 7 tabs in total.

Examples of each tab are shown below:

Survival Days



Impact of Management Action	Before	After
Survival days	0	366
Shortfall date	01 Oct 2022	> 01 Oct 2023
Lowest Point in 1 year	-136,143	
Lowest Point Date	27 Feb 2023	
Lowest Point day	D150	

Inflows and Outflows (before and after stress impact) - Table

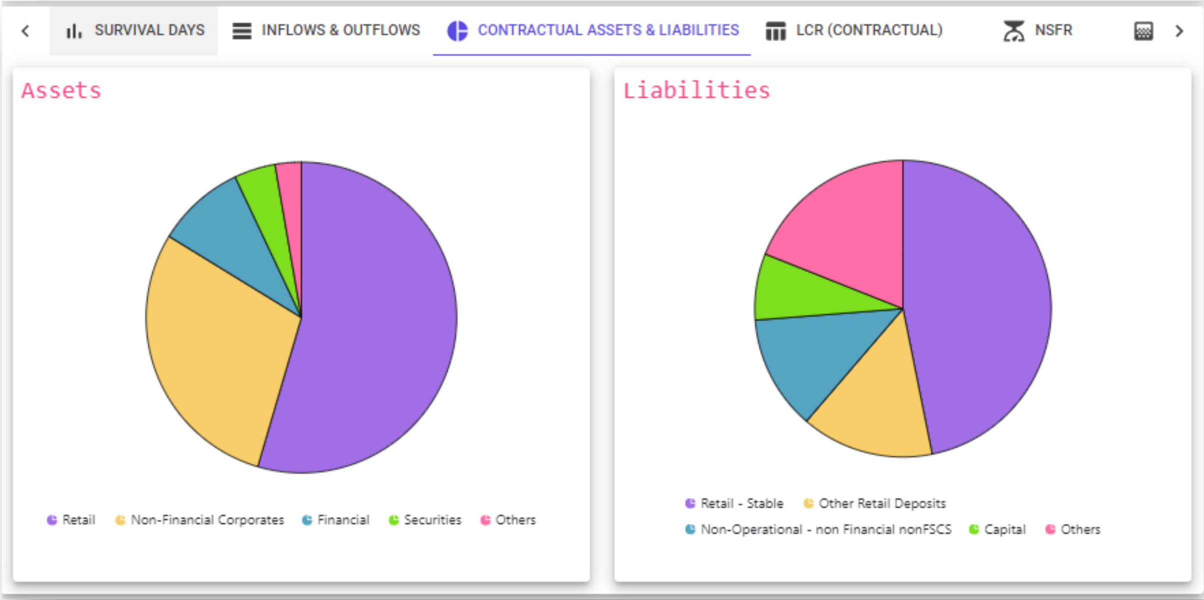
LCR | SURVIVAL DAYS | INFLOWS & OUTFLOWS | CONTRACTUAL ASSETS & LIABILITIES | LCR (CONTRACTUAL)

Inflows	Before	%	Stress Impact	After	%
Monies due from retail customers	1,683,489	54.54%	-63,010	1,620,479	60.75%
Monies due from non-financial customers not corresponding to principal repayment	3,956	0.13%	-1,978	1,978	0.07%
Monies due from non-financial corporates	901,539	29.21%	-118,952	782,587	29.34%
Monies due from financial customers	284,907	9.23%	-259,907	25,000	0.94%
Monies due from securities maturing	132,556	4.29%	-62,341	70,215	2.63%
Other inflows	3,099	0.10%	89,422	92,521	3.47%
Other HQLA Assets (excl. securities)	76,927	2.49%	-30,672	46,255	1.73%
Total assets	3,086,473	100.00%	-418,799	2,667,674	100.00%

Outflows	Before	%	Stress Impact	After	%
Capital	186,600	7.24%	-1,870	184,730	8.56%
Deposits subject to higher outflows - category 1	103,067	4.00%	-25,810	77,257	3.58%
Deposits subject to higher outflows - category 2	92,496	3.59%	-30,975	61,521	2.85%
Stable deposits	1,206,931	46.84%	-91,263	1,115,667	51.71%
Other retail deposits	372,178	14.45%	-62,754	309,424	14.34%
Non-Operational deposits by financial-customers (incl. correspondent banking)	93,958	3.65%	-36,168	57,790	2.68%
Non-Operational deposits by non-financial customers not covered by FSCS	321,606	12.48%	-93,926	227,680	10.55%
Liabilities arising from operating expenses	177,220	6.88%	-53,636	123,584	5.73%
Total liabilities	2,576,452	100.00%	-418,799	2,157,653	100.00%
Off-balance sheet net outflows	91,690	3.56%	-89,422	2,267	0.11%

Wholesale Funding / Ratio	592,784	23.01%	-183,730	409,054	18.96%
Retail Customer Deposits / Retail Customer Funding Ratio	1,797,067	69.75%	-233,199	1,563,869	72.48%
Capital Funding / Ratio	186,600	7.24%	-1,870	184,730	8.56%
Total funding	2,576,452	100.00%	-418,799	2,157,653	100.00%

Inflows and Outflows (Contractual i.e. before stress) – Chart Analysis



LCR (Contractual i.e. before stress)

INFLOWS & OUTFLOWS | CONTRACTUAL ASSETS & LIABILITIES | LCR (CONTRACTUAL) | NSFR | MISMATCH GAP

Liquidity Coverage Ratio	Contractual
Total outflows	160,391
Total inflows	291,209
Inflows after 75% cap (if applicable)	120,293
Net Liquidity outflow	40,098
Liquidity Buffer	196,141
Liquidity Cover Ratio (Pillar I)	489.16%
Pillar 2 Buffer Add-on	5,200
Net liquidity output (Pillar 1 + Pillar 2)	45,298
Liquidity Buffer	196,141
Liquidity Cover Ratio (Pillar I + Pillar II)	433.00%

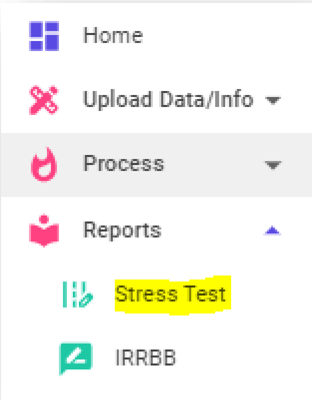
NSFR (Summary results before and after stress)

Net Stable Funding Ratio	Contractual	Stress
Available Stable Funding (ASF)	2,033,649	1,771,753
Required Stable Funding (RSF)	1,297,320	1,304,606
Net Stable Funding Ratio	156.76%	135.81%

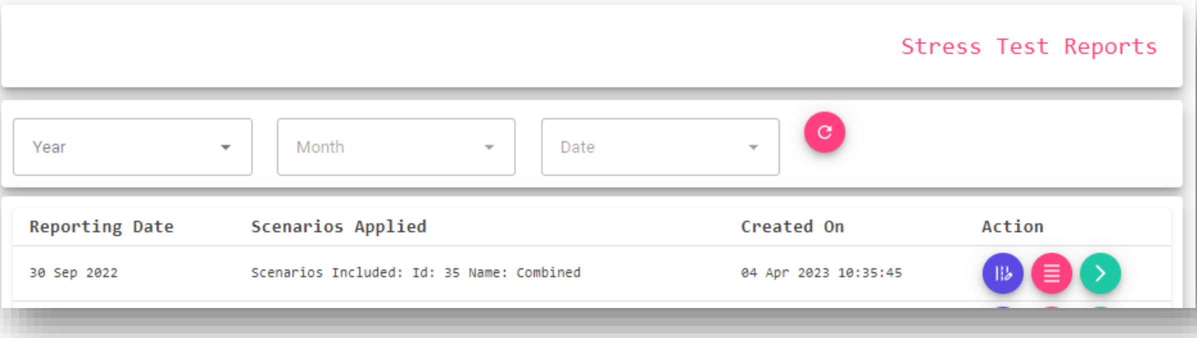
Mismatch Gap (Time bucket Analysis of Cashflows and Overall Action taken)

Mismatch gap (refinancing requirement)	< 14 days	< 30 days	< 3 mths	< 6 mths	< 9 mths	< 1 year	Overall
Contractual	-619,999	-720,612	-1,062,507	-1,260,702	-1,395,387	-1,443,482	
Stress (before management action)	-139	20,726	-117,104	-111,587	-100,021	-62,504	
Management Action (utilising CBC)	88,461	0	38,541	9,141	0	0	136,143
Stress (after management action)	88,322	109,187	9,898	24,556	36,122	73,639	
Additional Management Action (rebuild CBC)	0	0	0	25,000	15,000	5,000	45,000
Post Management Action (incl rebuilding CBC)	88,322	109,187	9,898	49,556	76,122	118,639	Total : GBP 181,143

12)In order to allow further analysis and to support wider distribution, the full reports can be downloaded into Microsoft Excel. To do this, navigate to the navigation bar and click on **Reports -→ Stress Test.**



13) The following screen will be shown, providing all the Stress testing reports, by reporting date, scenario and creation date.



Under the Action Column, the user can select one of the following options:

Purple Button – Click this to download the Microsoft Excel Report.

Pink Button – Click this to download the Microsoft Excel Datamaster (see Section 3.3).

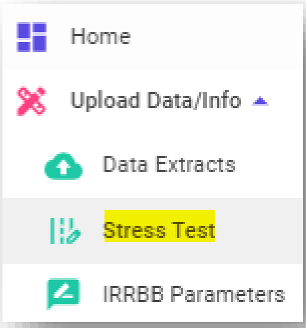
Green Button – Click this to re-access the dashboards on the k-ALM interface (as shown in step 12).

3 Additional functions

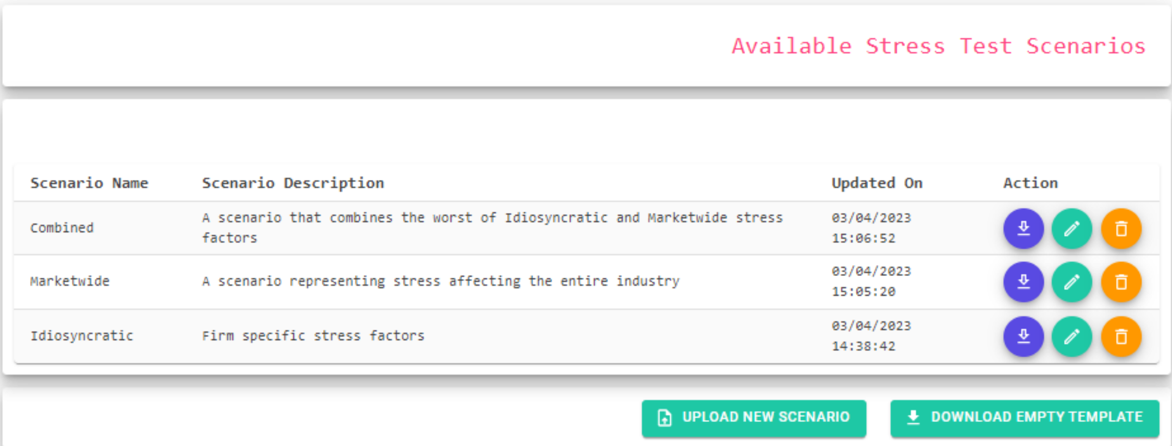
3.1 Creating a new scenario

A new stress scenario can be created as-and-when required. The steps to create a new scenario is given below.

1) To begin the process, Navigate to **Upload Data/Info → Stress Test** on the left hand-side menu select

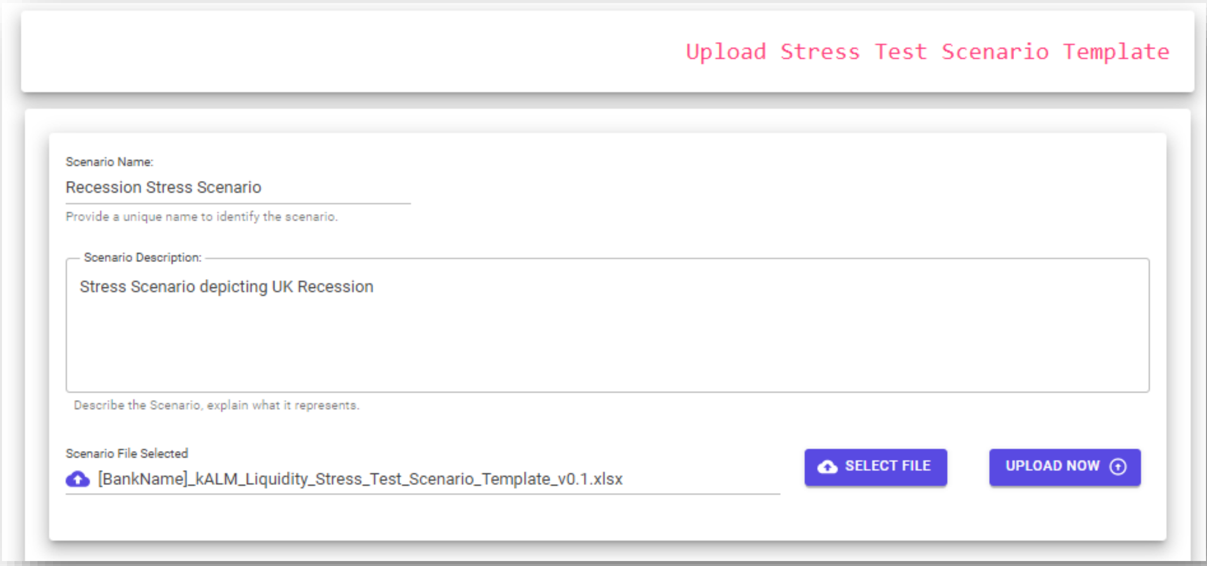


The following screen will be show:

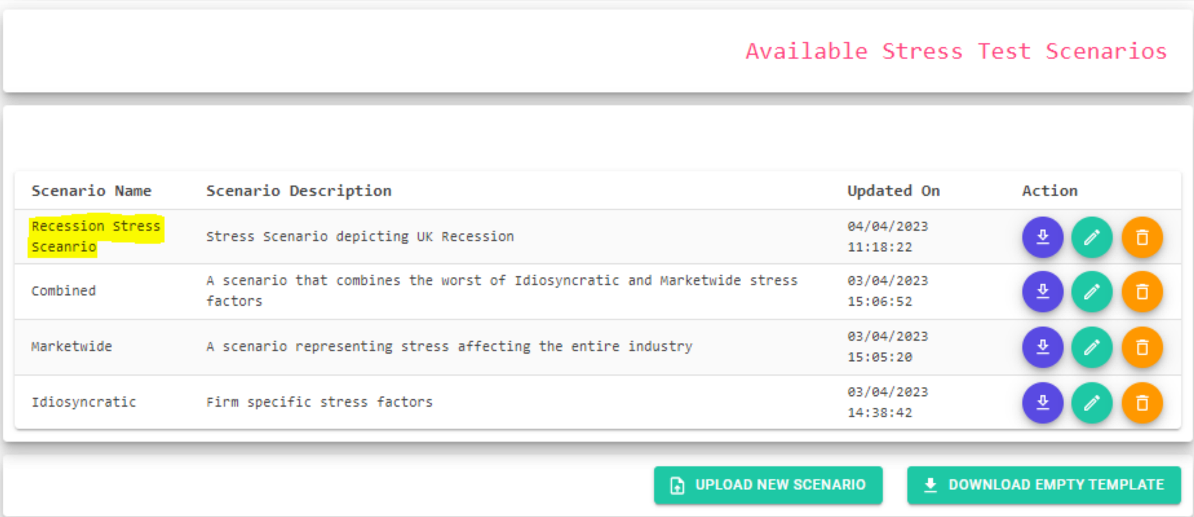


2) If the user is applying a completely new scenario, Click on "DOWNLOAD EMPTY TEMPLATE". This will download a Microsoft Excel 'Liquidity Stress Assumption' Template onto the users local environment, which will need to be updated. Instructions on how to update this sheet is provided in the Functional Coverage section of this User Manual under Section 4.

3) Once completed, click on "UPLOAD NEW SCENARIO" and enter a new scenario name, description and select the file containing the new stress scenario and then Click on "UPLOAD NOW"



4) Once processed, the new scenario will be created and will be shown along with the other existing scenarios, found under Reports → Stress Test



On this screen, under the Action Column, the user can also select one of the following options, if required:

Purple Button – Click this to download the Microsoft Excel Scenario Template.

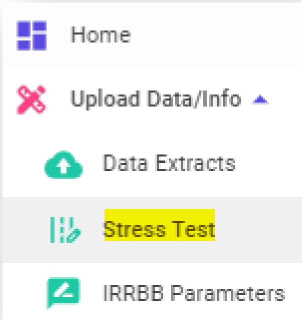
Green Button – Click this to edit the existing scenario (see section 3.2).

Orange Button – Click this to delete the Stress Sceanrio.

3.2 Edit an existing scenario

As part of the k-ALM solution, the model provides three prescribed stress scenarios (Idiosyncratic, Market-wide and Combined). In the event, the user wishes to modify a stress scenario (prescribed or any new stress scenario), the user can do this by following the process detailed below:

1) To begin the process, Navigate to **Upload Data/Info → Stress Test** on the left hand-side menu select

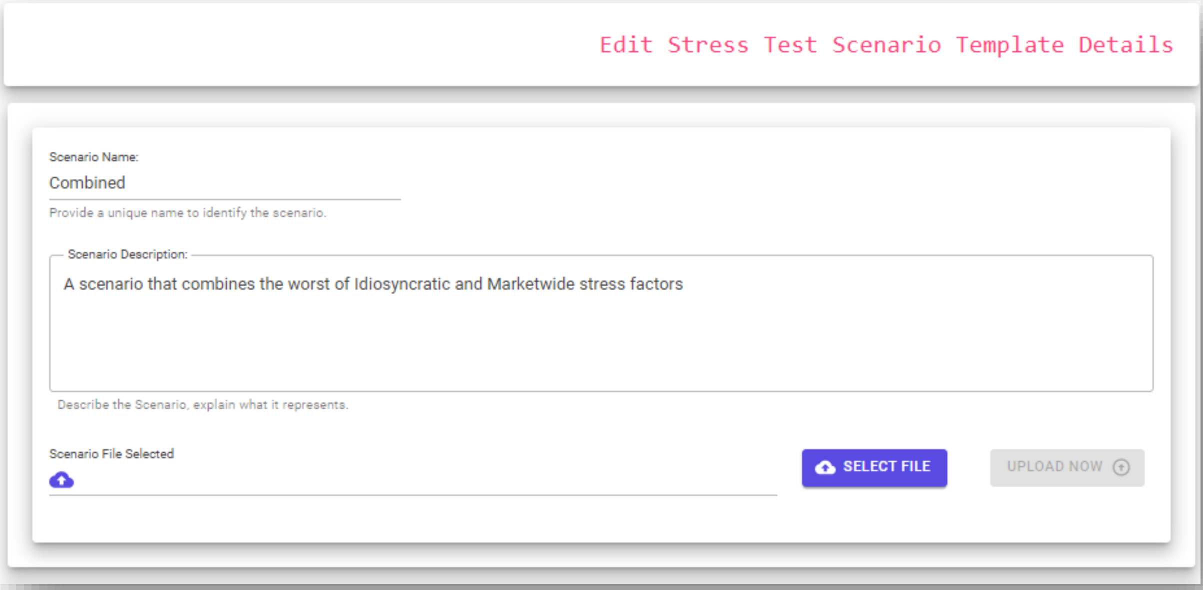


The following screen will be shown:

2) Click on the **Purple Button** of the desired stress scenario to download the scenario. A Microsoft Excel file will be downloaded onto the users local environment.

3) Update the excel with the modified stress assumptions.

4) Once complete, navigate to the Stress Scenario page, as detailed in Point 1 of this section. Click on the **Green Button** (i.e. Edit Scenario).



5) Click on "SELECT FILE" and open the updated Stress Scenario Template. Modify the Scenario name and Description as required, and then click on "UPLOAD NOW".

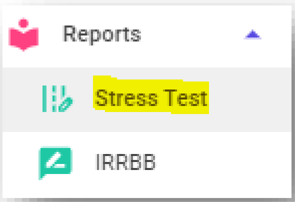
   Once completed the scenario will now be updated.

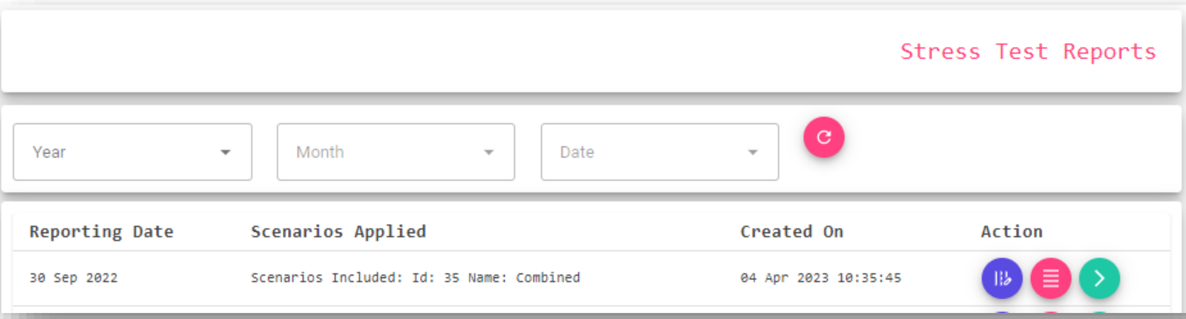
   Follow the steps described in Section 2.2 to execute the stress scenario.

3.3 Detailed data download

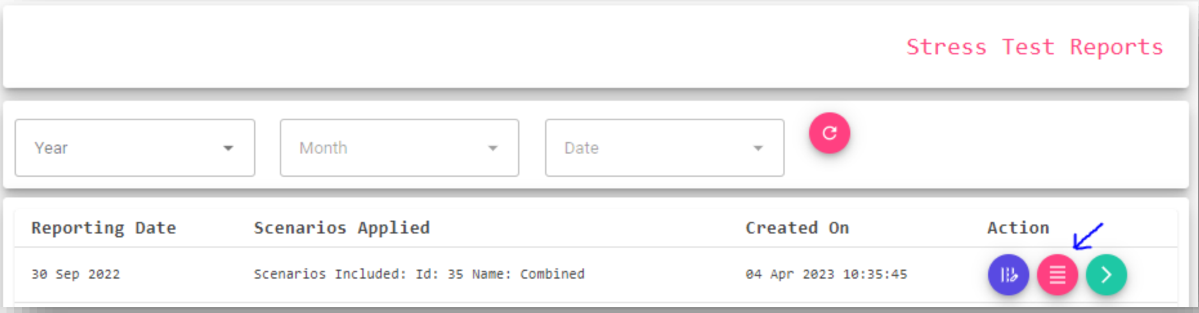
To download the detailed data of the stress for a given date, use the following steps:

1) On the left-hand menu, , navigate to **Reports → Stress Test**





2) Identify the appropriate reporting date and click on the **Pink button** under the Action Column (i.e., download datamaster).



Once completed, the datamaster will download on to the user's local environment, in Microsoft Excel format.

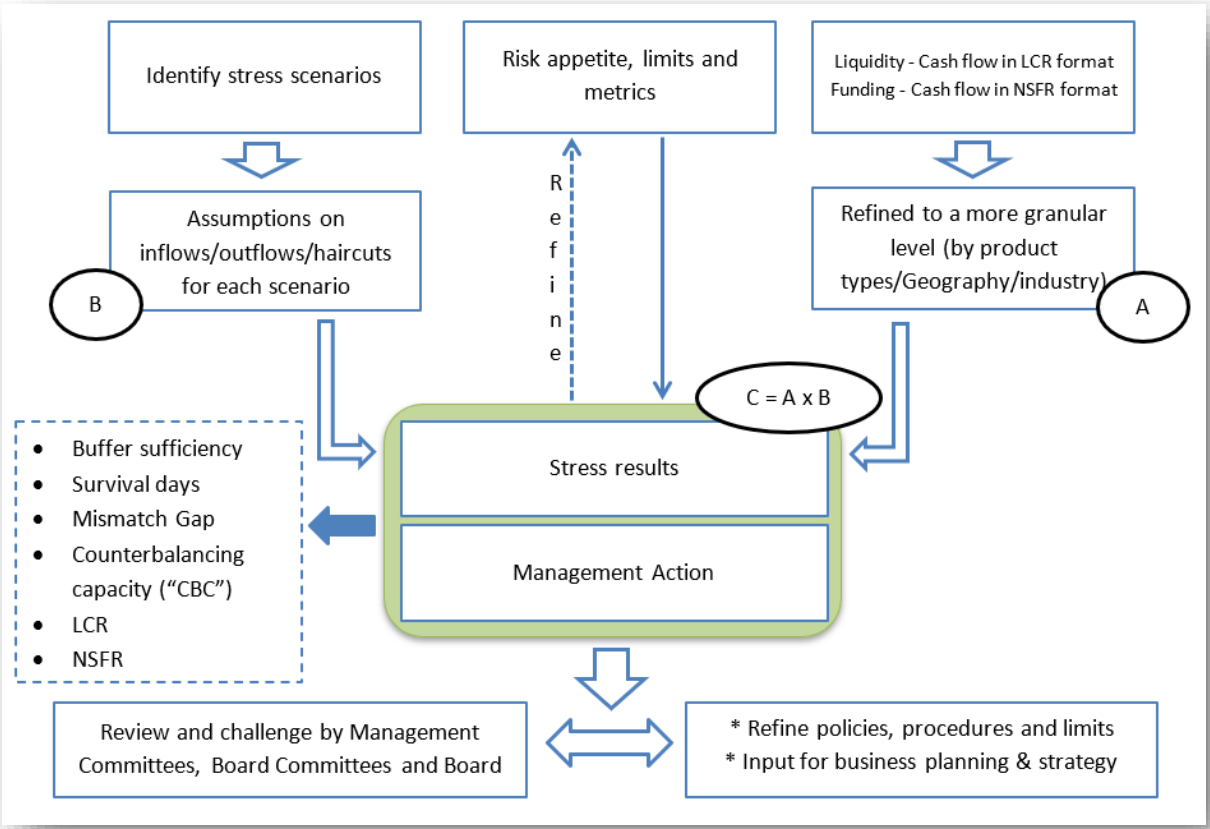
FUNCTIONAL COVERAGE

1 Introduction

This document sets out the approach and inner workings of the k-ALM Liquidity stress testing tool. The tool is developed by Katalysys Ltd and provides banks with a one stop Liquidity and Funding assessment tool, which allow users to apply stress scenarios to any financial position and assess the analyse the impact of liquidity risk based on various scenarios over a one-year period.

The tool and resulting outputs are designed to be practical and understandable by any user and provide a more robust liquidity risk management process that can be used as part of the bank's ongoing risk management monitoring process and annual Internal Liquidity Adequacy Assessment Process ("ILAAP").

A very highly level summary of the liquidity and Funding process from start to finish is provided below:

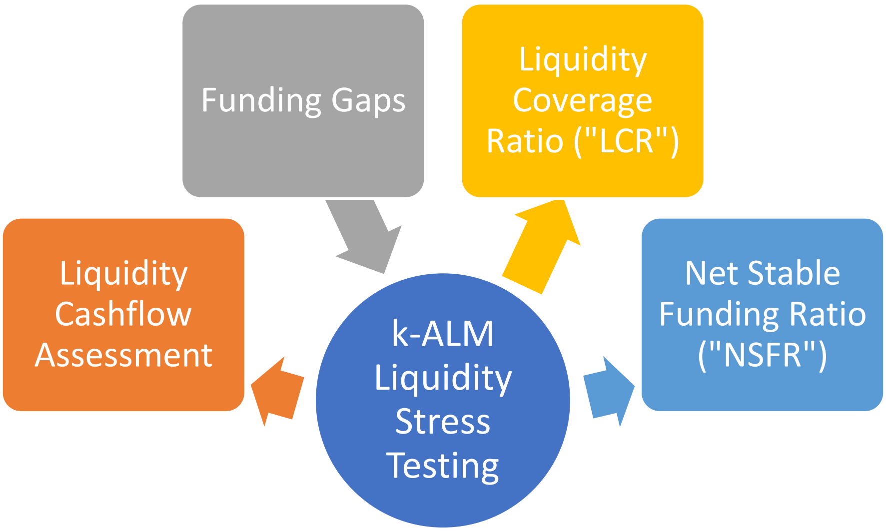


As illustrated above, the process is also designed to help provide senior management with the necessary information to make informed decisions regarding future risk limits, risk policies & procedures and the overall business strategy.

2 The Tool

2.1 Key outputs

The tool is designed to generate three fundamental outputs based on scenario testing:



Scenario testing is an important functionality provided by the k-ALM tool, in which the user has the flexibility, to analyse the impact on various liquidity metrics, based on the scenario transactions entered. The tool will generate a fully detailed analysis of liquidity and funding results, which will allow key users to conduct a more in depth analysis and make an informed decision on risk management limits, policies and procedures.

The Liquidity cashflow assessment will be used to identify the peak liquidity shortfall (i.e. refinancing requirement) over a user specific daily period (i.e. any number of days up to 365 days) and a fixed 1 year period. The highest shortfall identified will then be used as part of the management actions assessment. In addition, the same analysis will be used to assess the LCR at different periods over 1 year. This includes an LCR at the end of Day 14, Day 30, Month 3, Month 6, Month 9 and 1 Year.

The NSFR assessment, which is used to analyse the stability of a bank's funding profile, provides an assessment of the NSFR under stress conditions, which a user can assess to ensure the bank continues to hold enough stable funding to cover the duration of the long-term assets.

Following the completion of all assessments, a comprehensive summary will also be generated alongside the main results, which will help provide a clear overview of the stress results, which is designed to be easily distributed and understood by the wider members of the Bank and circulated within internal circles/committees.

2.2 Data Inputs

In order for the tool to produce such outputs, the tool requires a number of inputs as of a given reference date. Inputs are largely from the Bank's internal liquidity regulatory reporting systems which are used to generate the current relevant returns such as the PRA110, LCR and NSFR.

Dependent on the bank's systems and data files, the bank would only be required to upload a number of input sheets, before the k-ALM tool processes these files and generates the outputs.

A summary of the data required for each of the output is detailed below:

Liquidity cashflows and LCR

The liquidity cashflow model is created using a combination of the underlying regulatory reporting data used to generate the LCR and PRA110. Since the LCR only looks at the initial 30 days, the PRA110 is also used to further expand the dataset and provide the liquidity inflows and outflows for the life of each lending product and deposit.

This dataset includes all cashflows (including future interest) as per contractual terms, across various maturity periods. This includes:

1) Bank assets that create a cash flow (e.g. loans)
2) Bank liabilities that create a cash flow (e.g. deposits)
3) Bank's off-balance sheet commitments, which could potentially give rise to cash outflows
4) Bank's operating expenses

The combination of these two data sets will be key for generating both the liquidity cashflows and resulting LCR results.

Net Stable Funding Ratio ("NSFR")

The NSFR is an accompanying output that will mirror the liquidity stress assumptions and apply these on the contractual funding position. The data will again make use of the underlying regulatory reporting data used to generate the NSFR returns.

Compared to the liquidity cashflows, this assessment requires the use of the:

1) the accounting value for all items, including accrued interest, except for derivatives.
2) For derivatives positions in the same netting set, it is the fair value of the positions on a net basis
3) For derivatives positions that do not belong to a netting set, it is the fair value of the positions on a gross basis

Ultimately, since accounting values are used this dataset should reconcile back to the Bank's balance sheet based on the given reference date.

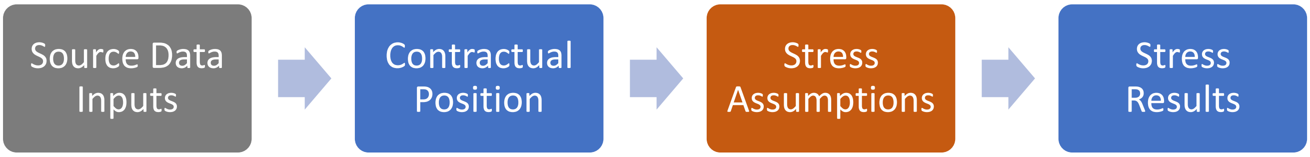
2.3 Model Layout

The model is laid out in a number of reports which are shown below:

Sections	Output Section Names
Liquidity Stress testing	Liquidity Contractual
	Liquidity Stress Assumptions
	Liquidity Stress Results (incl. management actions)
Funding Stress testing	Funding Contractual
	Funding Results
Liquidity and Funding Summary	Liquidity and Funding Summary

As shown above, there are 2 key sections to the model: **1) Liquidity** and **2) Funding**.

For both Liquidity and Funding, the model is further split in 3 sub-key components: The **Contractual Position, Stress Assumptions** and **Stress Results**. A summarised flow of information from inputs to results are shown below:



The model is designed to be an end-to-end automated process, however, the one key input required by the user would be within 'Liquidity Stress Assumptions', which allow the user to implement their own stress conditions and parameters.

The following sections will provide an overview of each element, which will expand on the key components and the areas where user inputs will be required.

Liquidity Stress Testing

3 Liquidity Contractual

3.1 Time Buckets

The 'Liquidity Contractual' report displays the Bank's entire liquidity position (i.e. cashflows at a given reference date). This is displayed both at a time bucket level (i.e. a summarised bucket which encapsulates a certain number of days for example B01 = Up to 14 days) and a daily time bucket level out to 1 year.

The contractual sets out 10-time buckets which are shown below:

B00	B01	B02	B03	B04	B05	B06	B07	B08
Non-defined maturity	Up to 14 days	Up to 30 days	Up to 3 months	Up to 6 months	Up to 9 months	Up to 1 year	Up to 2 years	Up to 5 years

As for the daily time bucket analysis, the model set out the cashflows for each day up to 1 year (i.e. 365 or 366 days). A small snippet is shown below:

D001	D002	D003	D004	D005	D006	D007
01 Apr 2022	02 Apr 2022	03 Apr 2022	04 Apr 2022	05 Apr 2022	06 Apr 2022	07 Apr 2022

All products are mapped to the individual time buckets and daily buckets and are derived based on the maturity date and reference date. For example: if the model reference date was 31st December 2021 and a loan of £5m is maturing on 31st May 2022. Therefore, the residual maturity would be calculated as 31/05/2022 minus 31/12/2021 = 151 days. Based on a residual maturity of 151 days, the loan would be mapped to time bucket B04 (up to 6 months) and Daily bucket D151.

One key difference between the two is that the summarised time buckets will capture the liquidity cashflows for the entirety of each item life, as compared to the daily buckets which will only look at a 1 year period.

3.2 Row Classification

The contractual will display all applicable items in rows based on the LCR classifications (i.e. Row used in LCR templates C72 to C74).

The rows are splits into 6 key sections:

LCR Classification Sections	Contents
Retail Outflows	Deposits from retail customers e.g. Current Accounts, Saving Accounts, Term Deposits etc.
Retail Inflows	Monies due from retail customers e.g. mortgages, personal loans etc.
Wholesale Outflows	Deposits from corporates and financial customers e.g. Current Accounts, Saving Accounts, Term Deposits etc.
Wholesale Inflows	Monies due to customers other than retail e.g. corporates, multilateral development banks, Sovereign, financial customers etc. products could

LCR Classification Sections	Contents
	include mortgages, business loans, nostro balances, money market placements etc. In addition to the loan products, this section would also include monies due from securites maturing i.e. Nominal value of LCR and non-LCR qualifying securities.
Off-Balance Sheet Items	This section would include all off-balance items e.g. credit lines, trade finance deals, overdrafts pipeline loans etc. In addition, inflows and outflows from derivatives will also be included as part of this section.
Marketable Assets and Reserves	Includes the market value of the Bank's securities (Both LCR and non-LCR qualifying securities) that are considered marketable, and other reserves such as cash and central bank reserves.

A snapshot of a selection of rows within the wholesale Inflows section are shown below:

	WHOLESALE INFLOWS(%)
C74.0040	Monies due from non-financial customers not corresponding to principal repayment
C74.0070	Monies due from non-financial corporates
C74.0080/90	Monies due from sovereigns, MDBS and PSE, and other legal entities
C74.0120-130	Monies due from financial customers being classified as operational deposits
C74.0150	monies due from central banks not being classified as operational deposits
C74.0160	Monies due from financial customers
C74.0180	Monies due from trade financing transactions
C74.0190	Monies due from securities maturing

The level shown in the figure above is at its highest, however, the model will also allow users to assess the impact of a liquidity stress at a more detailed sub-categorisation level, which may be based on product categories, Industry classification, geographical locations or even a bespoke level (as specified by the Bank). For each line, the Bank has the option to use up to nine sub-categories with the tenth being earmarked as 'other'.

An example of sub-categorisation for 'monies due from financial customers' is shown below, where six sub-categories have been used:

C74.0160	Monies due from financial customers
C74.0160.0	*Money Market placements (excl. Group)*
C74.0160.1	*Monies due from financial entities*
C74.0160.2	*Monies due from Group entities (Nostro and Placements)*

User Note: The model has been designed to account for all applicable rows that small- and medium-sized Banks may require. However, in instances where a specific LCR row is not initially available, the model has been designed to incorporate a number of empty rows (each with ten sub-categories), that can be used as and when required.

3.3 Cumulative Mismatch

Upon completion of the contractual data inputs and row classification, the contractual flows will be automatically generated along with the Cumulative Mismatch, which takes into account the Net Outflows and Net Inflows based on both the summarised time buckets and daily time buckets. An example of the time bucket-wise mismatch is shown:

ALL Currencies	B00	B01	B02	B03	B04	B05	B06	B07	B08	B09
Row Description	Non-defined maturity	Up to 14 days	Up to 30 days	Up to 3 months	Up to 6 months	Up to 9 months	Up to 1 year	Up to 2 years	Up to 5 years	Over 5 years
	31 Mar 2022	14 Apr 2022	30 Apr 2022	30 Jun 2022	30 Sep 2022	31 Dec 2022	31 Mar 2023	31 Mar 2024	31 Mar 2027	31 Dec 9999
Net outflow (Retail + Wholesale + Off-balance sheet)	(25,000,000)	-	(200,000)	(600,000)	(500,000)	(600,000)	(6,000,000)	(20,000,000)	-	-
Net inflow (Retail + Wholesale + Off-balance sheet)	2,500,000	3,000,000	150,000	35,000,000	25,000,000	37,000,000	30,000,000	80,000,000	100,000,000	25,000,000
CUMULATIVE MISMATCH (Retail + Wholesale + Off-balance sheet)	(22,500,000)	(19,500,000)	(19,550,000)	14,850,000	39,350,000	75,750,000	99,750,000	159,750,000	259,750,000	284,750,000

3.4 Inflow/Assets and Outflow/Liabilities Tables

Following an assessment of the cumulative mismatch, the model also generates an unbalanced 'Assets and Liabilities' table. These tables are based on the same cashflows used to generate the contractual liquidity position I.e. Total inflows are considered Assets and total outflows are considered Liabilities.

These tables are constructed to demonstrate the change in the composition of the Bank's balance sheet, following a stress event and the impact of subsequent management actions. This comparison is detailed as part of the 'Liquidity Stress Results' section, which can be found in Section 5.

Assets		
Monies due from retail customers	-	0.00%
Monies due from non-financial customers (except for central banks) not corresponding to principal repayment	-	0.00%
Monies due from non-financial corporates	194,591,422	43.52%
Monies due from sovereigns, multilateral development banks and PSE	23,736,213	5.31%
Monies due from other legal entities	-	0.00%
Monies due from financial customers being classified as operational deposits where the credit institution is able to establish a corresponding symmetrical inflow rate	-	0.00%
Monies due from financial customers being classified as operational deposits where the credit institution is not able to establish a corresponding symmetrical inflow rate	-	0.00%
Monies due from financial customers	33,148,421	7.41%
Monies due from trade financing transactions	-	0.00%
Monies due from securities maturing	127,102,881	28.43%
Other inflows	68,551,865	15.33%
Inflows from secured lending and capital market-driven transactions - collateral that qualifies as a liquid asset	-	0.00%
Inflows from secured lending and capital market-driven transactions - collateral that does not qualify as a liquid asset	-	0.00%
Other HQLA Assets (excl. securities)	-	0.00%
Empty Inflow	-	0.00%
Empty Inflow	-	0.00%
Empty Inflow	-	0.00%
Empty Inflow	-	0.00%
Total assets	**447,130,804**	**100.00%**

It should be noted that, the 'Assets and Liabilities' tables, will **not** balance each other and this is expected. This is because the model uses cashflows which include future interest.

3.5 LCR

As part of the 'Liquidity Contractual' section, the model will also calculate the LCR at the given reference date. This is generated based on the daily cashflows and applicable LCR weight. As shown below, the model will calculate both the Pillar 1 LCR and Pillar 1 + Pillar 2 LCR.

Liquidity Coverage Ratio	
Total outflows	1,600,000
Total inflows	5,000,000
Inflows after 75% cap (if applicable)	1,200,000
Net liquidity outflow	400,000
Liquidity Buffer	3,000,000
Liquidity Coverage Ratio (Pillar 1)	**750.00%**
Pillar 2 Buffer Add-On	500,000
Net liquidity outflow (Pillar 1 + Pillar 2)	900,000
Liquidity Buffer	3,000,000
Liquidity Coverage Ratio (Pillar 1 + Pillar 2)	**333.33%**
Current regulatory LCR requirement	100%
Excess buffer available after meeting Pillar I & Pillar 2	**2,100,000**

3.6 Key Reconciliation Points

As part of the Contractual cash flows, the model has an in-built reconciliation check which is used to ensure the 1 year daily cashflows reconcile back to the sum of the time buckets up to 1 year (i.e. Sum of B01 to B07).

When successful, the reconciliation check, at the end of the daily cashflows will display a tick sign, as shown below:

Reconciled

Not Reconcile

In the event, the check displays an '**X**', the 1 year cashflows will need to be checked to ensure the cashflows are corrected.

4 Liquidity Stress Assumptions

The 'liquidity Stress Assumptions' is a key input that is required to be updated by the user.

This section includes the following elements:

- Stress assumptions (i.e. Inflow/Outflow percentages for each stress condition) for all items reported in the contractual flows.
- Haircuts applicable to marketable assets.
- Daily stress period Parameter (i.e. Identification of number of days to assess daily liquidity cashflows within 1 year)
- Management Actions – Proportion of Marketable assets available for Repo
- Management Actions - Additional lending applied to ensure franchise viability (only if not already considered within initial inflow assumptions)
- Management Actions – Rebuilding of HQLA buffer from realisation of non-HQLA Assets.
- Additional Actions – Bank's ability to raise new capital or new funding
- Risk Appetite – Setting of the risk appetite limits for the LCR and NSFR

Each bullet point is described in more detail below:

4.1 Stress Assumptions

4.1.1 Application of Assumptions under time buckets

The cash inflows and outflows under each of the stress conditions are expressed as a percentage of the contractual position.

- For items that do not have a defined maturity (e.g. current accounts), the inflow or outflow is assumed to happen on day one.
- For items with defined maturity, (e.g. term deposits), the percentage determines the inflow or outflow on maturity of the deal/contract.

For example, if the liability is for £1,000 and the outflow assumption is 20%, then the outflow due to stress is calculated as £1000 x 20% = £200. In other words, 20% of the liability is withdrawn and 80% is rolled over.

Based on this approach, depending on the stress scenario determined by the Bank, for each applicable line item, a stress assumption will need to be recorded in this section. As previously mentioned, the Bank has the option to select a predefined stress scenario, which will be included as part of the k-ALM tool kit or add in their own bespoke stress assumptions using a Blank Template (See Section 3.1 of the Technical Coverage Section).

Adding in a bespoke stress assumption is very straightforward can be completed by updating the percentages of each relevant line item and time bucket. An example of the application is shown below for 'Monies due from financial customers':

Stress Assumptions	B00	B01	B02	B03	B04	B05	B06	B07	B08	B09
Row Description	Non-defined maturity	Up to 14 days	Up to 30 days	Up to 3 months	Up to 6 months	Up to 9 months	Up to 1 year	Up to 2 years	Up to 5 years	Over 5 years
Monies due from financial customers										
Money Market placements (excl. Group)	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Monies due from financial entities	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Monies due from Group entities (Nostro and Placements)	100%	0%	0%	0%	0%	0%	0%	100%	100%	100%

It should be noted that the tool performs the stress over a 1 year period only. Any amount falling in the greater than 1 year bucket is determined to be unstressed. Hence in the figure shown above in time buckets (B07 to B09 – to the right of the black solid line), the stress assumption applied is 100% (i.e. Full repayment expected). Similarly, for Outflows, the stress assumption of 0% is applied (i.e. No expectation that any liability will be withdrawn and 100% will be rolled over).

4.1.2 Early termination of deposits

The Bank also has the option to opt-in for early termination of deposits (Retail or Wholesale), which could be used in a stress scenario where there is a severe run on the Bank and retail customers have lost faith in the Bank's ability to safeguard its funds. In this case, the Bank can apply the relevant percentage in the final column. The resulting outflow will be spread across 2 periods (i.e. 50% of early terminations on day 15 and the remaining 50% on day 31), and the amount terminated will be calculated based on the total contractual outflows exceeding 3 months (i.e. total exposures within time buckets B04 to B09).

In a simplified example, assuming an early termination of 10%, if the total retail outflows were £10m of which £3m was within 30 days, the exposure terminated early would be an additional £700,000 (i.e. total outflows exceeding 3 months of £7m multiped by 10%).

Deposits subject to higher outflows – category 1								Early termination %
Deposits subject to higher outflows – category 1	30%	30%	30%	30%	15%	15%	15%	20.00%

The figure above shows where the early termination assumptions are applied (Figure coloured in red).

4.1.3 Ongoing Gradual Withdrawals

The additional element relating to stress assumptions is the ability to apply an assumption relating to the gradual withdrawal of deposits (Retail or Wholesale) of Non-maturing deposits (i.e. Current accounts/Saving Accounts/ISAs etc).

For example, if the bank wishes to gradually spread the outflow of current account deposits, the user can do this by spreading the initial stress assumption out over the desired periods. E.g. as shown below, assuming the original stress assumption was 30%, this has been spread out over the 6 months.

Row Description	Non-defined maturity	Up to 14 days	Up to 30 days	Up to 3 months	Up to 6 months	Up to 9 months	Up to 1 year		
Deposits subject to higher outflows – category 1								Early termination %	Ongoing gradual withdrawals
Current accounts	0.00%	15.00%	7.50%	3.75%	3.75%	0%	0%		Y
Savings accounts & Instant access accounts	0.00%	15.00%	7.50%	3.75%	3.75%	0%	0%		Y

The user will also be required to enter "Y" in the final column to ensure the process takes this into account.

User Note: Ongoing withdrawal should ONLY be applied for lines that exclusively have only non-defined-maturity items (e.g., current accounts, savings accounts). In circumstances where this is required, a breakdown by deposit type will be required.

4.1.4 Reliance on short-term funding

In the event the Bank is reliant on short-term funding (e.g., overnight money market borrowing), then the residual rollover amounts from the previous time-bands should be considered as the starting position for the calculation of LCR for the next time-band. An example of this is shown in Section 5 (see LCR). In such instances, and additional flag can be set to "Y" to ensure that the rollover amounts are considered for the LCR calculation.

	C73.0040.9	Other			Reliant on short term funding?
x	C73.0060	Deposits subject to higher outflows - category 1	Early termination %	Ongoing gradual withdrawls?	Y

4.2 Haircuts on Marketable Assets

One of the model's key management actions is the sale of marketable assets, which could be conducted to raise liquidity within a short timeframe.

However, under stress conditions, it would be highly unlikely that such assets could be realised at their current market value, hence a haircut should be applied to demonstrate the realistic value that could be raised in the event such action is required to be implemented. These haircuts are to be applied in the following table:

	MARKETABLE ASSETS & RESERVES	Haircut/Fall in Value (Due to Stress)	% that can be Repo'ed	Realisation feasible?
C72.0040	Level 1 - Coins and banknotes	0.00%		
C72.0050	Level 1 - Withdrawable central bank reserves	0.00%		
C72.0060-130	Level 1 - Central bank, central government assets, MDB and similar (CQS1)	2.00%	0.00%	
C72.0140	Level 1 - Qualifying CIUs shares/units (underlying is Level 1 assets)	2.00%	0.00%	
C72.0190	Level 1 - Others (e.g., qualifying Covered bonds)	2.00%	0.00%	
C72.0240-250	Level 2A - Central bank/Government or Public Sector Entity assets (RW20%)	15.00%	0.00%	Y
C72.0280	Level 2A - Corporate debt securities (CQS1)	15.00%	0.00%	Y
C72.0290-300	Level 2A - Others	15.00%	0.00%	Y
C72.0360-380	Level 2B - Corporate debt securities (CQS2/3) and shares (major stock ind	50.00%	0.00%	Y
C72.0390-470	Level 2B - Others	50.00%	0.00%	Y
C72.9009	Non-LCR qualifying marketable assets			
C72.9009.0	Monies due from (non-LCR Investment grade) securities	55.00%	0.00%	Y
C72.9009.1	Monies due from (non-LCR non-investment grade) securities	75.00%	0.00%	Y

As shown above, there are a number of additional columns: 1) *haircut/Fall in value (Due to stress)*, 2) *% that can be Repo'ed and 3) Realisation Feasible*. For this section, Points 1 and 3 will be covered here. Refer to Section 4.4 for Point 2.

The first column – Haircut/Fall in Value (Due to stress), is based on the realisation of LCR qualifying marketable assets or the non-LCR qualifying marketable assets and is used to demonstrate the sale of marketable assets in order to cover any liquidity shortfall at the earliest point.

The third column – Realisation feasible, allows the bank to identify if the specific portfolio should be realised under stress conditions. An environment where the user would enter an "N" would be where the haircut is too high and would involve a significant profit and loss impact. As part of the model, level 1 Assets are defaulted to "Y", as it is unlikely that haircuts on such securities would lead to material capital impacts.

In addition, as part of the model, the Bank has the ability to rebuild the HQLA buffer by realising residual Non-LCR qualifying assets (after initial realisation to cover shortfalls) and using the funds to rebuild the HQLA buffer.

4.3 Daily Stress Period within 1 year

As previously mentioned, the liquidity stress model analyses both the cashflows on a daily timeframe and on a time-bucketed timeframe. However, depending on a Bank's risk management approach to liquidity, a differing range of timeframes may need to be modelled. As a result, this model will allow a user to determine the number of days it wishes to assess the peak liquidity shortfall. To adjust such requirements, a parameter (shown in yellow) in the 'liquidity stress assumption' template can be updated to the user's desired day range. **We recommend setting this to 365.**

Daily stress period within 1 year	
Number of days assessed (up to 366 days)	92

4.4 Management Action – Repurchase Agreements ("Repo")

The second set of management actions would be the use of the Repo markets. If Repo lines are available to the Bank, a selection of marketable assets could be repo'ed to raise additional funds in a very short timeframe.

As part of the model, the user should update the second column '% that can be Repo'ed' - this is shown below:

MARKETABLE ASSETS & RESERVES	Haircut/Fall in Value (Due to Stress)	% that can be Repo'ed	Realisation feasible?
Level 1 - Coins and banknotes	0.00%		
Level 1 - Withdrawable central bank reserves	0.00%		
Level 1 - Central bank, central government assets, MDB and similar (CQS1)	2.00%	0.00%	
Level 1 - Qualifying CIUs shares/units (underlying is Level 1 assets)	2.00%	0.00%	
Level 1 - Others (e.g., qualifying Covered bonds)	2.00%	0.00%	
Level 2A - Central bank/Government or Public Sector Entity assets (RW20%)	15.00%	100.00%	Y
Level 2A - Corporate debt securities (CQS1)	15.00%	100.00%	Y
Level 2A - Others	15.00%	100.00%	Y
Level 2B - Corporate debt securities (CQS2/3) and shares (major stock index)	50.00%	100.00%	Y
Level 2B - Others	50.00%	100.00%	Y
Non-LCR qualifying marketable assets			
Monies due from (non-LCR Investment grade) securities	55.00%	100.00%	Y
Monies due from (non-LCR non-investment grade) securities	75.00%	0.00%	Y

The "% that can be Repo'ed' will determine the percentage of the portfolio that is available to be repo'd. For example, under stress assuming a portfolio contains securities of £50 million, only 80% may be available for repo. Therefore the action would take £40 million into consideration.

If the Bank has no repo lines, the '% that can be Repo'ed' should remain at 0%, which indicates that the specific portfolio is not available for repo.

4.5 Management Action – Additional Lending

As part of management actions, the model allows users to apply additional lending (i.e. Inflows) conducted to ensure franchise viability. This action is designed to ensure funds can be continued to be lent ensuring the bank's franchise availability remains intact during the stress period. In the table shown below, the user can apply a percentage of the current inflows amounts, for example if the stress, models inflows of £10 million after stress relating to monies due from non-financial corporates. Assuming the user applies a 10% assumption, the amount of additional lending applied would be £1 million. This should only be used in instances where the inflows assumptions do not factor in an element of re-lending.

Additional lending for franchise viability (ONLY if not already considered as inflow assumption	
Additional lending to ensure franchise viability on exiting stress	**% of current**
Monies due from non-financial corporates	0%
Monies due from sovereigns, MDBS and PSE, and other legal entities	0%
Monies due from financial customers being classified as operational deposits	0%
Monies due from central banks not being classified as operational deposits	0%
Monies due from financial customers	0%
Monies due from trade financing transactions	0%
Other inflows	0%
Inflows from secured lending and capital market-driven transactions	0%
Other bank specific inflows	0%
Monies due from retail customers	0%
Other Inflows (for later use)	0%
Other Inflows (for later use)	0%

4.6 Management Action – Additional Action - Raising new capital or Funding

Where necessary, the bank may be required to take further action to order to raise new funding or capital, in order to cover any residual liquidity shortfall or to rebuild the banks' HQLA buffer.

Within this model the user has a number of options that it can apply including:

1) Securitisation
2) Sale of other assets
3) Raise additional retail deposits (by increasing rates)
4) Raise Capital
5) Other borrowings (wholesale)
6) Enforce higher rates to certain assets to reduce rollover (i.e., enforce earlier repayment)
7) Others

The user can apply these actions in the table shown below:

Management Action - New Capital/Funding	Non-defined	Up to 14 days	Up to 30 days	Up to 3 months	Up to 6 months
Securitisation					
Sale of other assets					
Raise additional retail deposits (by increasing rates)					
Raise Capital					
Other borrowings (wholesale)					
Enforce higher rates to certain assets to reduce rollover (i.e., enforce earlier repayment)					9,000
Others					
Total additional funding		-	-	-	9,000.00

In the example above, £9 million of new funding is being raised within the 6 month period. The time buckets where action can be applied is the same as that used in the contractual basis (i.e. Non-defined up to Over 5 years).

4.7 Risk Appetite Limits – LCR, NSFR and Survival Days

The final input, required by the user is the LCR, NSFR and Survival days risk limits as set out in the Bank's risk appetite statement.

This is used throughout the model, to visually show what stage the ratios are at, in relation the risk appetite.

An example of the input sheet is provided below:

Risk Appetite - LCR	Green	Amber	Red
LCR %	200.00%	160.00%	125.00%
NSFR %	130.00%	130.00%	120.00%
Survival Days (in days)	120	92	91

In the event, survival days is not considered applicable to the Bank, 91 days should remain as default.

5 Liquidity Stress Results

The 'Liquidity Stress Results' is the final aspect that brings together both the 'Liquidity Contractual' and 'Liquidity Stress Assumptions'. The layout of this output is very similar to the contractual with the summarised time bucket and daily analysis layout.

However, as expected, this includes a lot more detail to fully show the impact of the liquidity stress, the peak liquidity shortfall, the use of management actions, the change in Balance sheet composition due to stress and the impact on the LCR over differing periods.

This section will run through all the key elements of the stress results, which can be seen in the downloaded stress results report (See Section 2.2 of the Technical Coverage Section for instructions on how to download the Microsoft Excel report).

5.1 Stressed liquidity flows and Peak Shortfall

The stressed inflows are derived based on the contractual flows and stress assumptions, for each line item that is applicable to the Bank. This section is fully automated based on the initial inputs and stress assumptions applied by the user.

Similar to the contractual, the model will regenerate the cumulative mismatch for each time bucket and for each day up to 1 year. The exact same approach, as detailed in Section 3.3, is taken.

One of the key elements of this section is the identification of the Peak liquidity Shortfall. The model identifies the peak shortfall over a 1 year period including the CBC and one excluding the CBC using the cumulative mismatch analysis. Survival days are also so shown which indicates the point at which net cumulative cashflows turn negative during the 1 year daily analysis.

An example of the analysis is shown below, where £5 million was the peak shortfall identified in the 365 day analysis, the exact day and date of the peak shortfall are also presented here:

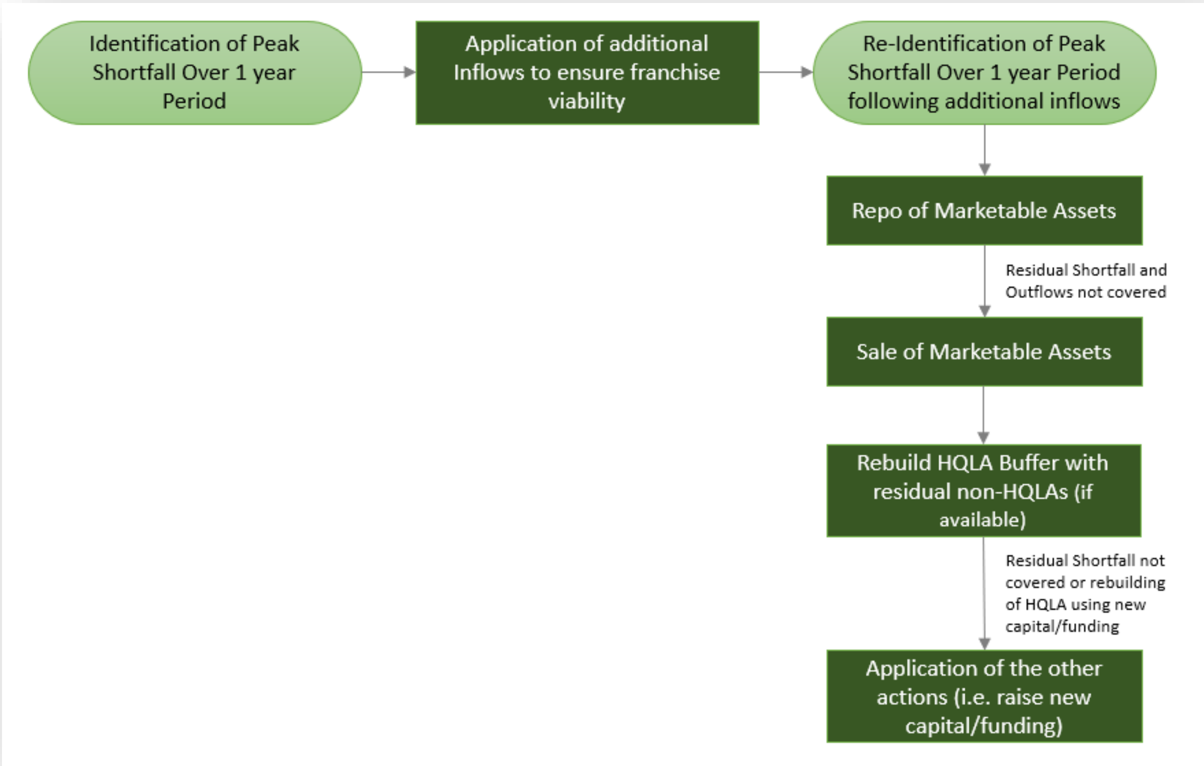
Management action requirement (peak deficit identification)		**Date**	**Day**
Highest (shortfall) over 365 days (daily analysis) excl. CBC	(5,000,000)	27-Feb-2023	D150
Highest (shortfall) over 365 days (daily analysis) incl. CBC	60,000	27-Feb-2023	D150
Management action required for (peak shortfall)	**(5,000,000)**		

Based on the example shown above, the peak shortfall of £5 million will then be used for assessing the amount and type of management actions applied. i.e. Shortfall of £5 million will be covered using management actions.

5.2 Management Actions

Management actions are used into model to cover the Peak Shortfall identified over a 1 year period.

The use and order of management actions are presented in the flow chart below:



As demonstrated above, the model is designed to ensure all shortfalls are sufficiently covered along with the rebuilding of the HQLA buffer, to ensure an LCR greater than the Board risk appetite (specified in Section 4.7) in the 1 year period.

The following sub-sections provide an overview of each management action and the assumption applied (if any).

5.2.1 Additional Lending – Franchise Viability Action

As part of management actions, the model allows users to apply additional learning conducted to ensure franchise viability. This action is designed to ensure funds can be continued to be lent out ensuring the bank's franchise viability remains intact during the stress period. As noted in Section 4.5, the user can apply a percentage of the existing inflows as additional new lending. If any additional lending is applied, the model will take these inflows into account and re-calculate the cumulative mismatch, along with the peak liquidity shortfall both on the user-specified daily analysis (up to 366 days) and the time bucket analysis. This feature should only be used in exceptional circumstances, as the inflow assumptions would have already factored in the additional lending from funds received.

Additional lending for franchise viability	Additional %												
Monies due from non-financial corporates	20%	(500)	(500)	(2,000)	(9,000)	(5,500)	(8,000)				(3,000)	(25,500)	(25,500)
Monies due from sovereigns, MDBS and PSE, and other legal entities	0%	-	-	-	-	-	-				-	-	-
Monies due from financial customers being classified as operational deposits	0%	-	-	-	-	-	-				-	-	-
Monies due from central banks not being classified as operational deposits	0%	-	-	-	-	-	-				-	-	-
Monies due from financial customers	0%	-	-	-	-	-	-				-	-	-
Monies due from trade financing transactions	0%	-	-	-	-	-	-				-	-	-
Other inflows	0%	-	-	-	-	-	-				-	-	-
Inflows from secured lending and capital market-driven transactions	0%	-	-	-	-	-	-				-	-	-
Other bank specific inflows	0%	-	-	-	-	-	-				-	-	-
Monies due from retail customers	0%	-	-	-	-	-	-				-	-	-
Other Inflows (for later use)	0%										-	-	-
Other Inflows (for later use)	0%										-	-	-
Total additional outflows		(500)	(500)	(2,000)	(9,000)	(5,500)	(8,000)				(3,000)	(25,500)	(25,500)
Net outflow (Retail + Wholesale + Off-balance sheet)	(20,000)	(100,000)	(25,000)	(170,000)	(40,000)	(30,000)	(30,000)	-	-	(400,000)	(410,000)	(530,000)	(940,000)
Net inflow (Retail + Wholesale + Off-balance sheet)	110,000	95,000	48,000	20,000	55,000	52,000	60,000	4,000	300,000	2,400,000	280,000	450,000	3,000,000
CUMULATIVE MISMATCH (Retail + Wholesale + Off-balance sheet)	90,000	85,000	108,000	(42,000)	(27,000)	(5,000)	25,000	29,000	329,000	2,329,000			

Management action requirement (peak deficit identification) - including additional act		Date	Day
Highest shortfall over 365 days (daily analysis)	(130,000)	27-Feb-2023	D150
Highest refinancing gap over a 1 year period (timeband analysis)	(42,000)		
Management action required for (peak shortfall)	(130,000)		

In this example, the user has applied a 20% increase of additional inflows over a 1 year period which equates to £25.5 million. Once applied the new cumulative mismatch is calculated along with a new Peak shortfall, of which additional management actions will be applied to cover the shortfall.

5.2.2 Repo of Marketable Assets

A key management action built into the model is the repo of marketable assets. This would be applicable to all Banks, where repo lines are available. As mentioned in Section 4.4, the user will be required to set the percentage of assets available to repo, within the 'Liquidity Stress Assumption' template.

If this management action is available, the model assumes that funds raised from the repo of LCR qualify assets can be raised from day 1, however, the model will apply the management action as and when required based on a forward looking requirement of the first 14 days and then every 3 months. This will continue to be applied until the relevant portfolio has been fully utilised. Funds from the repo of non-LCR qualifying assets will be received after 5 days (i.e. Day 005). Even though this option would enable a Bank to raise funds in a very short period of time, due to the nature of repos, funds will need to be repaid and this is assumed to take place after 1 year (i.e. outside the stress period).

5.2.3 Sale of Marketable Assets

The most crucial management action built into the model is the sale of marketable assets. As detailed in Section 4.2, the user will be required to set the haircut rate (due to stress) and feasibility flag for each relevant asset portfolio, within the 'Liquidity Stress Assumption' template.

Similar to the Repo, the realisation of the marketable assets will occur as and when required based on the 1 year daily cashflows. The model will initially identify a forward looking 14 day requirement and will apply the necessary action to ensure any negative mismatch (shortfalls) are fully covered via the sale of marketable assets. This process is replicated every 3 months and allow the bank to illustrate a more realistic stress event of when the buffer is required and when action is taken. This will further help the analysis of the movement in the LCR over a 1 year period.

An example of a sale of marketable assets is detailed below:

Realisation of CBC (via drawdown, sale or Repo)	Total % utilised									Total utilised	Haircut	Excess available	
								Shortfall before reaslisation of CBC and assets		(137,000)			
Level 1 - Withdrawable central bank reserves	100.00%	75,000	-	-	-	-	-	75,000	75,000	75,000	-	-	
Level 1 - Central bank, central government assets, MDB and similar (CQS1)	64.45%	12,000	-	40,000	10,000	-	-	52,000	62,000	62,000	(2,000)	34,110	
Level 1 - Qualifying CIUs shares/units (underlying is Level 1 assets)	0.00%	-	-	-	-	-	-	-	-	-	-	-	
Total Funding raised by Sale of Assets		87,000	-	40,000	10,000	-	-	127,000	137,000	137,000	(2,000)	64,110	
Cumulative realisation impact of CBC		87,000	87,000	127,000	137,000	137,000	137,000				(2,000)	<===Direct P&L Impact	
Cumulative mismatch after utilising CBC		172,000	195,000	85,000	110,000	132,000	162,000						

In this example, a residual shortfall of £137 million was identified of which 100% of the Level 1 Central Bank Reserves and 64.5% of other Level 1 assets were realised in order to cover the shortfall. It should be noted, the sale of assets will also result in the Bank incurring a haircut, which in this example was £2 million, which will reduce non-dated capital as it will have a direct Profit and Loss impact.

User Note: it should be noted that if repo's are available to the bank, no direct Profit and Loss impact will be incurred.

Once this action is concluded, the model will identify the residual shortfall required to meet the daily flows. However, in addition, the model will also assess the HQLA requirement to ensure the LCR can meet the Board Risk Appetite Limit (As set in by the user in the 'Liquidity stress assumption' template, see Section 4.7). Based on the higher requirements, the model will proceed to take additional action to cover this requirement. An example of this is shown below:

		To meet Daily flows for 365 days	To meet Board LCR limit of > 200.%
Shortfall after CBC realisation daily flows and to meet Board LCR Risk appetite	(66,000)	-	(66,000)

As shown above, there is no need for any additional action in relation to meeting the daily flows over a 1 year however, in order to ensure the LCR meets the Board limit of 200%, additional action of £66 million, would be required. Therefore £66 million will be required to rebuild the HQLA buffer. Actions specific to this are shown in the next two subsections.

5.2.4 Rebuilding of HQLA buffer

Upon completion of the sale of marketable assets, the Bank's HQLA buffer may have realized to a degree which could result in a breach of the LCR requirement (both Internal limits and regulatory).

In order to ensure an LCR greater than the Board risk limit, after management actions, the model incorporates a mechanism to rebuild the HQLA buffer to meet the Board risk limit, at a minimum, by realising any residual non-LCR qualifying marketable assets and using the resulting funds to purchase Level 1 assets.

Since only non-LCR qualifying assets are used to rebuild the buffer, the model assumes the funds will only be available after 3 months (i.e. Day 92), and funds raised will be dependent on the value the assets can be realised, after the sale of marketable assets to cover any liquidity shortfall, and application of haircuts (see Section 4.2 for applying the haircut)

In this example, the figure below demonstrates the conversion of Non-LCR qualifying assets to cover a shortfall of £8 million. As a result, 21.7% of the residual non-LCR qualifying marketable assets (high-quality) was realised after incurring a haircut of £1.2 million (as set by the user – see Section 4.2).

Assets available of Sale (Non-LCR qualifying assets)	%remaining utilised					Haircut	Shortfall
Non-LCR qualifying marketable assets				8,000,000			8,000,000
High Quality	21.72%			8,000,000		(1,200,000)	-
Non-High Quality	0.00%			-		-	-
Total Funding raised by Sale of Assets				8,000,000		(1,200,000)	

In the event, there are no residual non-LCR qualifying marketable assets, following the sale of marketable assets, the remaining option to rebuild the HQLA buffer will be via an injection of new capital/funding or other additional actions. This is detailed in the next section.

5.2.5 Other Additional Actions - Raise Capital or Funding

In the event, a residual requirement is still needed to rebuild the HQLA back up to meet the Board risk limit, the user can apply additional actions as detailed in Section 4.6.

Any input of new action, as conducted by the user, will be displayed in this part of the 'liquidity stress test' report, as shown below.

							Amount required to rebuild HQLA to Board Limits	(8,000)
							Total	Shortfall
Securitisation	-	-	-	-	-	-	-	(8,000)
Sale of other assets	-	-	-	-	-	-	-	(8,000)
Raise additional retail deposits (by increasing rates)	-	-	-	-	-	-	-	(8,000)
Raise Capital	-	-	-	-	-	-	-	(8,000)
Other borrowings (wholesale)	-	-	-	-	-	-	-	(8,000)
Enforce higher rates to certain assets to reduce rollover (i.e., enforce earlier repayment)	-	-	-	8,000	-	8,000	8,000	-
Others	-	-	-	-	-	-	-	-
Total funds raised	-	-	-	8,000	-	8,000	8,000	

As demonstrated above, £8 million of inflows arising from earlier repayments has been applied after 6 months. This inflow will adequately cover the residual requirement, ensuring the LCR will increase above the Board risk limit in the later period of the stress (i.e. after 6 months).

5.3 Inflows/Assets and Outflows/Liabilities

As previously discussed in Section 3.4, the model generates an unbalanced Assets and liabilities tables using the liquidity cash flows.

The tables in this section go one step further by applying the impacts from the stress and the corresponding actions taken, generating a before and after comparison, allowing users to reflect on the change in portfolios.

The figure below shows a sample of the Assets table and the change due to stress and management actions:

Inflows	Before		Stress effect	Actions	After	
Monies due from retail customers	1,600,000	53.28%	(64,000)		1,536,000	57.57%
Monies due from non-financial customers not corresponding to principal repayment	40,000	1.33%	(2,000)		38,000	1.42%
Monies due from non-financial corporates	700,000	23.31%	(100,000)		600,000	22.49%
Monies due from financial customers	250,000	8.33%	(200,000)		50,000	1.87%
Monies due from securities maturing	100,000	3.33%	(57,000)	-	43,000	1.61%

As demonstrated above, the table shows the change in each portfolio due to the impact of the stress ('Stress effect' column) and management actions ('Actions Column). For example, securities will reduce following the 'sale of marketable securities'.

5.4 LCR

In order to evaluate the LCR under stress, this model calculates the LCR at different stages of the stress up to 1 year. This includes an LCR on Day 1 and then at the end of Day 30, Month 3, Month 6, Month 9 and 1 Year.

The evaluation of the LCR is shown at 3 differing stages:

1) Before any management action
2) After management actions (Use of CBC but before rebuilding HQLA Buffer)
3) After management action (after rebuilding HQLA Buffer).

An example of the LCR assessments are provided below:

LCR Before Management Actions

Liquidity Coverage Ratio (before Management actions to cover peak shortfall)	Own Stress assessment					
	On Day 1	At the end of 30 days	At the end of 3 months	At the end of 6 months	At the end of 9 months	At the end of 1 year
Total outflows	160,000	92,000	58,000	69,000	73,000	71,000
Total inflows	291,000	33,000	14,000	9,000	12,000	12,000
Inflows after 75% cap (if applicable)	120,300	33,000	14,000	9,000	12,000	12,300
Net liquidity outflow	40,100	59,000	44,000	60,000	61,000	58,700
Liquidity Buffer	196,100	189,600	189,600	189,600	189,600	189,600
Liquidity Coverage Ratio (Pillar 1)	489%	321%	431%	316%	311%	323%
Pillar 2 Buffer Add-On	5,200	5,200	5,200	5,200	5,200	5,200
Net liquidity outflow (Pillar 1 + Pillar 2)	45,300	64,200	49,200	65,200	66,200	63,900
Liquidity Buffer	196,100	189,600	189,600	189,600	189,600	189,600
Liquidity Coverage Ratio (Pillar 1 + Pillar 2)	433%	295%	385%	291%	286%	297%

LCR After Management Actions – after use of Counterbalancing Capacity

STAGE 1: Liquidity Coverage Ratio (After Management actions to meet all outflows)	Own Stress assessment					
	On Day 1	At the end of 30 days	At the end of 3 months	At the end of 6 months	At the end of 9 months	At the end of 1 year
Total outflows	160,000	92,000	58,000	69,000	73,000	71,000
Total inflows	291,000	33,000	14,000	9,000	12,000	13,000
Inflows after 75% cap (if applicable)	120,300	33,000	14,000	9,000	12,000	13,000
Net liquidity outflow	40,100	59,000	44,000	60,000	61,000	58,000
Liquidity Buffer	196,100	210,000	63,000	76,000	96,000	134,000
Liquidity Coverage Ratio (Pillar 1)	489%	356%	143%	127%	157%	231%
Pillar 2 Buffer Add-On	5,200	5,200	5,200	5,200	5,200	5,200
Net liquidity outflow (Pillar 1 + Pillar 2)	45,300	64,200	49,200	65,200	66,200	63,200
Liquidity Buffer	196,100	210,000	63,000	76,000	96,000	134,000
Liquidity Coverage Ratio (Pillar 1 + Pillar 2)	433%	327%	128%	117%	145%	212%

LCR After Management Actions – after use of Counterbalancing Capacity and including additional action taken to rebuild the HQLA Buffer

STAGE 2: Liquidity Coverage Ratio (After Management actions - including HQLA rebuilding)	Own Stress assessment					
	On Day 1	At the end of 30 days	At the end of 3 months	At the end of 6 months	At the end of 9 months	At the end of 1 year
Total outflows	160,000	92,000	58,000	69,000	73,000	71,000
Total inflows	291,000	33,000	14,000	9,000	12,000	13,000
Inflows after 75% cap (if applicable)	120,300	33,000	14,000	9,000	12,000	13,000
Net liquidity outflow	40,100	59,000	44,000	60,000	61,000	58,000
Liquidity Buffer	196,100	210,000	63,000	101,000	136,000	179,000
Liquidity Coverage Ratio (Pillar 1)	489%	356%	143%	168%	223%	309%
Pillar 2 Buffer Add-On	5,200	5,200	5,200	5,200	5,200	5,200
Net liquidity outflow (Pillar 1 + Pillar 2)	45,300	64,200	49,200	65,200	66,200	63,200
Liquidity Buffer	196,100	210,000	63,000	101,000	136,000	179,000
Liquidity Coverage Ratio (Pillar 1 + Pillar 2)	433%	327%	128%	155%	205%	283%

Overall, The Tables demonstrate the change in LCR at different stages of the stress. For example the Day 30 LCR looks at the stressed period from Day 1 to Day 30 and the unstressed period from Day 31 to Day 60 (i.e. 30 full days). This approach assumes that all remaining outflows and inflows after stress are rolled over on a daily basis. This allows the user to evaluate the LCR impact as time progresses over a stress period.

For the LCR outflow calculation on close of business on say day 30, the following calculation is done for each LCR line based on the stress assumptions:

(A) Include all deposits from the undefined maturity period that are not been withdrawn yet. For example, if there were £10 million deposits has undefined maturity, and 80% was withdrawn; then the remaining £2 million is considered)

(B) Include all deposits maturing from day 31 to day 60 (next 30 days) say £3 million

C = LCR outflow rate x [(A) + (B)] : The standard LCR factors (say 40%) are then applied to the total withdrawable deposits = 40% x [£2 m + £3 m]

A simplified example is provided below, as to how the outflows for the various period are calculated:

		Undefined maturity	**31 to 60 days**
A	Contractual outflows	£10,000	£3,000
B	Bank's own internal stress outflow assumption	80%	
C = A x B	Outflows due to stress	£8,000	
D = A - C	Residual balance still remaining	£2,000	
E	Residual balances to consider from previous time-bands		£2,000
F = E + A	Total outflows due to stress		£5,000
G	LCR standard outflow factor		40%
H = F x G	Total outflow for LCR		£2,000

Additional note: As part of the 'LCR after management actions – use of CBC', the LCR for the first 3 periods reduces down significantly before rising. This is due to sale of the LCR qualifying marketable assets in the initial stages of the stress period and the rebuilding of the buffer in the latter stages.

In the event the bank is reliant on short-term funding, and the appropriate flag is set in the assumptions (see Section 4 above – Short-term funding reliance) then even funds that are rolled over from the previous time-band is also considered in the LCR calculation. A simplified example is provided below, as to how the outflows for the various period are calculated:

		Day 1 to Day 30	**31 to 60 days**
A	Contractual outflows	£10,000	£3,000
B	Bank's own internal stress outflow assumption	80%	
C = A x B	Outflows due to stress	£8,000	
D = A - C	Residual balance still remaining	£2,000	
E	Residual balances to consider from previous time-bands		£2,000
F = E + A	Total outflows due to stress		£5,000
G	LCR standard outflow factor		40%
H = F x G	Total outflow for LCR		£2,000

5.5 Key Reconciliation Points

5.5.1 Time bucket vs Daily bucket cashflows

Similar to the 'Contractual' section, the 'liquidity Stress Results' has an in-built reconciliation checks which is used to ensure the 1 year daily stressed cashflows reconcile back to the sum of the stressed time buckets up to 1 year (i.e. Sum of B01 to B07).

When successful, the reconciliation check, at the end of the daily cashflows will display a tick sign, as shown below:

Reconciled

Not reconciled

There are similar reconciliation checks dotted throughout the report as shown above, these are mostly found within column NW of the downloaded Microsft Excel Report, for each separate management action.

The same checks are also conducted to check the cumulative mismatch (after management actions) at the end of the 3 months (i.e. Day 92) which can be found in Column DN of the downloaded Microsft Excel Report.

5.5.2 Assets vs Liabilities

As part of the Assets and Liabilities tables, a check has been added to ensure the difference between the Assets and liabilities before stress and after stress remains the same (i.e. zero). This is to confirm that the stress impacts and actions are equally applied to both sides of the balance sheet. This check is located to the right of the Inflows table.

An example of the check is shown below, the difference between the Assets and liabilities are equal, before and after stress:

A-L (Before)	A-L (After)
19,800,000	19,800,000
	-

Funding Stress Testing

6 Funding

6.1 Funding Contractual

The 'Funding Contractual', displays the Bank's entire funding position at a given reference date. The layout is far more simplified than liquidity due to the simplified nature of the regulatory NSFR returns.

The funding profile is set out in time buckets (i.e. a summarised bucket which encapsulates a certain number of days for example B01 = Up to 14 days) from B00 to B09 only.

The rows are based on the NSFR regulatory templates; C80 – required stable funding and C81 – Available stable funding. This ensures the correct NSFR factors are applied to the row item and provides a clear flow of information for the user.

A sample of the layout of the Contractual funding is shown below:

Ref.	FUNDING CONTRACTUAL - Marketwide : ALL Currencies : in GBP '000s Row Description
	AVAILABLE STABLE FUNDING
	Capital
C81.0030	Common Equity Tier 1
C81.0040	Additonal Tier 1
C81.0050	Tier 2
C81.0060	Other capital instruments
	ASF from retail deposits
C81.0090	Stable retail deposits
C81.0110	Other retail deposits
	ASF from other non-financial customers (except central banks)
C81.0160	*Liabilities provided by the central government of a Member State or a third country*
C81.0170	*Liabilities provided by regional governments or local authorities of a Member State or a third country*
C81.0180	*Liabilities provided by public sector entities of a Member State or a third country*
C81.0190	*Liabilities provided by multilateral development banks and international organisations*
C81.0200	*Liabilities provided by non-financial corporate customers*
C81.0210	*Liabilities provided by credit unions, personal investment companies and deposit brokers*

The 'Funding Contractual' levels are the same as that of the NSFR returns. Ultimately, the data used in the NSFR should reconcile back to the Balance sheet, where possible.

Once the contractual funding profile is complete, the model will automatically calculate the NSFR based on the relevant Required Stable Factors (RSF) and Available Stable Factors (ASF).

Total	390,000,000	312,000,000
NSFR = ASF / RSF	125.0%	

6.2 Funding Stress Assumptions

Where available the stress assumptions for the NSFR assessment will be based on the same conditions as recorded within the 'Liquidity Stress Assumptions' template. This ensures the stress is consistent across both assessments.

Ref.	FUNDING ASSUMPTIONS - Marketwide : ALL Currencies : in GBP '000s Row Description	B00 Non-defined maturity 31 Mar 2022	B01 Up to 14 days 14 Apr 2022	B02 Up to 30 days 30 Apr 2022	B03 Up to 3 months 30 Jun 2022	B04 Up to 6 months 30 Sep 2022	B05 Up to 9 months 31 Dec 2022	B06 Up to 1 year 31 Mar 2023	B07 Up to 2 years 31 Mar 2024	B08 Up to 5 years 31 Mar 2027	B09 Over 5 years 31 Dec 9999
	AVAILABLE STABLE FUNDING										
	Capital										
C81.0030	Common Equity Tier 1	100%									
C81.0040	Additonal Tier 1	100%									
C81.0050	Tier 2	100%									
C81.0060	Other capital instruments	100%									
	ASF from retail deposits										
C81.0090	Stable retail deposits	0%	0%	0%	0%	0%	0%	0%	100%	100%	100%
C81.0110	Other retail deposits	19%	33%	83%	25%	10%	8%	8%	100%	100%	100%
	ASF from other non-financial customers (except central banks)										
C81.0160	*Liabilities provided by the central government of a Member State or a third country*	100%	100%	100%	80%	80%	80%	80%	100%	100%	100%
C81.0170	*Liabilities provided by regional governments or local authorities of a Member State or a third country*	100%	100%	100%	80%	80%	80%	80%	100%	100%	100%
C81.0180	*Liabilities provided by public sector entities of a Member State or a third country*	100%	100%	100%	80%	80%	80%	80%	100%	100%	100%
C81.0190	*Liabilities provided by multilateral development banks and international organisations*	100%	100%	100%	80%	80%	80%	80%	100%	100%	100%
C81.0200	*Liabilities provided by non-financial corporate customers*	100%	100%	100%	80%	80%	80%	80%	100%	100%	100%
C81.0210	*Liabilities provided by credit unions, personal investment companies and deposit brokers*	100%	100%	100%	80%	80%	80%	80%	100%	100%	100%

6.3 Funding Stress Results

The 'Funding Stress Results' brings together both the 'Funding Contractual' and 'Funding Stress Assumption'. The layout mirrors the 'Funding Contractual' but the calculations are updated to apply the stress assumptions.

Similar to the liquidity approach, the results have a stress period of 1 year. Anything over 1 year is considered outside the stress period. This logic is also applied to the greater than 1 year ASF/RSF calculations, where the formulas only assess the value of required/available funding based on the contractual exposure (i.e. unstressed).

Once complete the stressed NSFR will be presented at the bottom of the report:

Total	**370,000,000**	**264,000,000**
NSFR = ASF / RSF	**140.2%**	

Liquidity and Funding Summary (Dashboard)

7 Summary

This summary report or dashboard provides users with an overview of the liquidity and funding position contractually (i.e. unstressed), under stress but before management actions and under stress after management actions.

The key components shown in the summary report include the LCR, NSFR, Mismatch Gap, Survival days and Assets & Liabilities Profile.

The 'Liquidity and Funding Summary' uses a variety of tables and charts to show the impacts of the stress conditions.

The tables and chart are shown and detailed in the following sub-sections.

User note: The tables and charts shown in this section will also be provided on the K-alm interface. (see Section 2.2 – point 12 of the Technical Coverage Section). There is no initial requirement to download the Microsoft Excel report to see the summary of results.

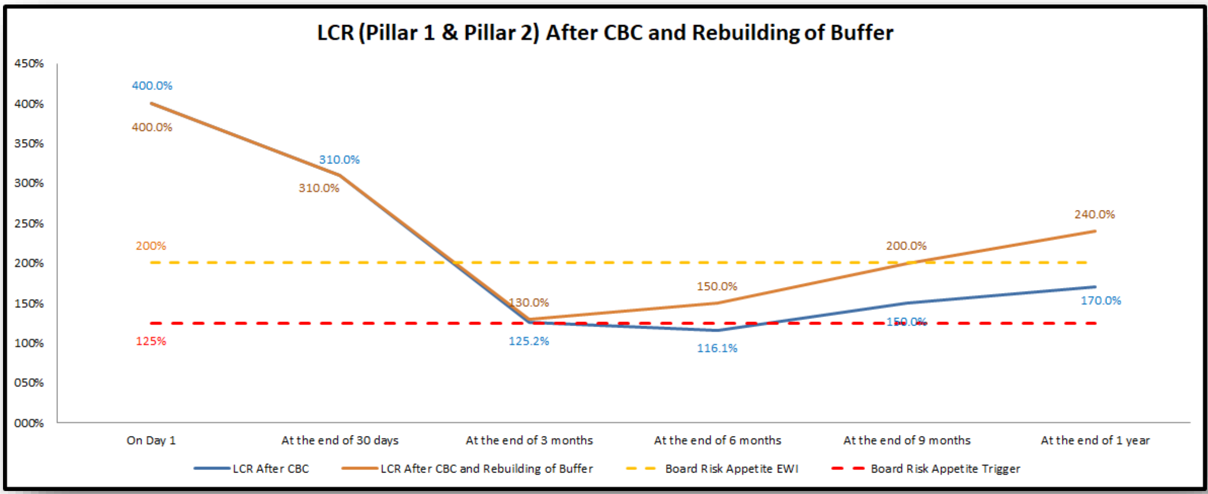
7.1 LCR

The summary report provides a complete analysis of the LCR at the unstressed position and at different time periods of the stress.

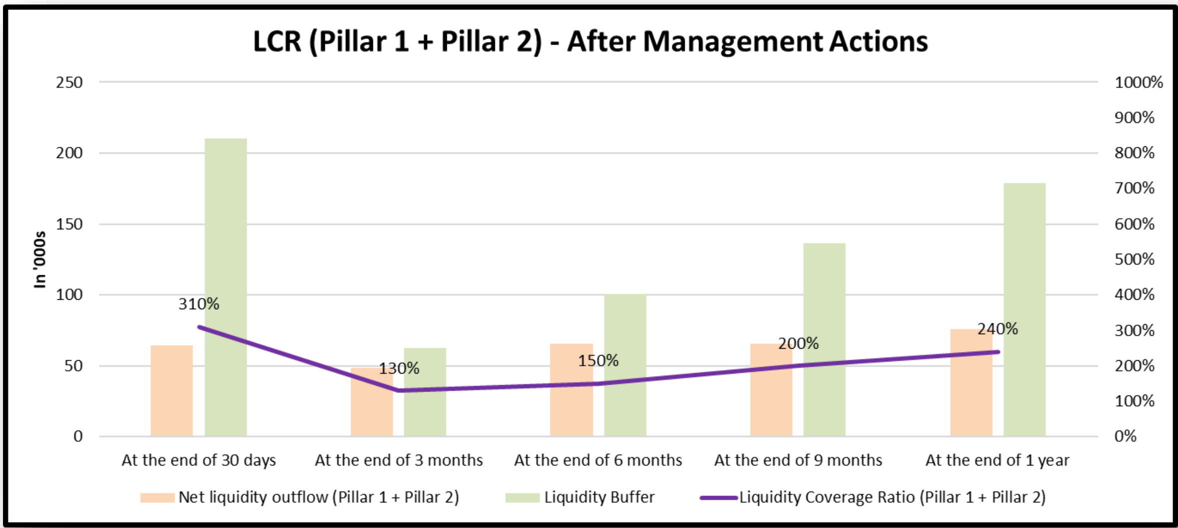
The unstressed position at the given stress reference date is provided in a table, as shown in the example below:

Liquidity Coverage Ratio	Contractual Assessment
Total outflows	160,000
Total inflows	300,000
Inflows after 75% cap (if applicable)	120,000
Net liquidity outflow	40,000
Liquidity Buffer	180,000
Liquidity Coverage Ratio (Pillar 1)	**450%**
Pillar 2 Buffer Add-On	5,200
Net liquidity outflow (Pillar 1 + Pillar 2)	45,200
Liquidity Buffer	180,000
Liquidity Coverage Ratio (Pillar 1 + Pillar 2)	**398%**

The LCR under stress at the different periods are captured in a chart as shown below. These periods include an LCR at the end of Day 30, Month 3, Month 6, Month 9 and 1 Year. This allows users to see the movement in LCR as the stress progresses and as management actions are exercised, the two lines shown in the chart represent the LCR after use of the CBC (Blue line) and then second being the LCR after use of the CBC and the rebuilding of the buffer (orange line), as demonstrated, the orange line shows that the action taken to rebuild the buffer was sufficient to be above the Board EWI limit by the end of the 1 year period:



The LCR under stress (after management actions – use of CBC and action to rebuild the buffer) is also provided in a chart format and provides a more detailed view including the net liquidity outflows and HQLA buffer level at each period. An example of the chart is provided below:



7.2 Mismatch Gap, Survival Days and Low Point

The Mismatch Gap (i.e. Refinancing Requirement) and Survival days are presented in the form of two tables and a chart.

The table for mismatch gap provides a clear image of where a refinancing requirement arises under specific time buckets. The table also provides the change in the mismatch gap based on 4 key stages:

1) The cumulative mismatch before stress (i.e. contractual unstressed position)
2) The cumulative mismatch after stress (no management actions applied)
3) The cumulative mismatch after stress (Management actions applied – Use of CBC only)
4) The cumulative mismatch after stress (Management actions applied – both use of CBC and rebuilding the Buffer)

It allows the user to identify where actions are required, where the action has been applied and to what amount, and the resultant mismatch gap. An example of such a table is shown below:

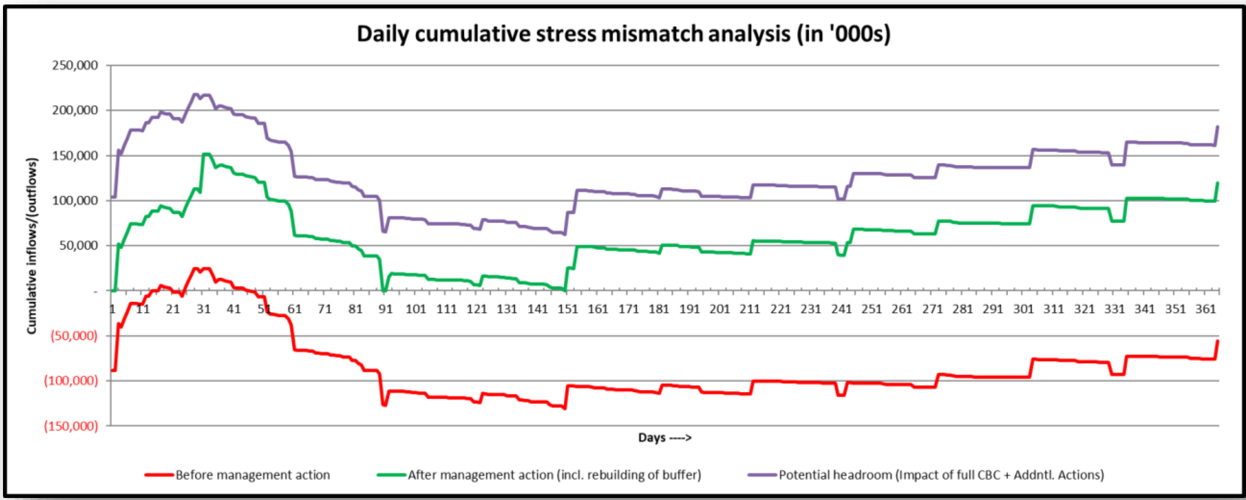
Mismatch gap (refinancing requirement)							Overall Management Actions Taken
	<=14 days	<=30 days	<=3 mths	<=6 mths	<=9 mths	<=1 year	
Contractual	(600,000)	(700,000)	(1,100,000)	(1,300,000)	(1,400,000)	(1,400,000)	
Stress (before management action)	-	20,000	(100,000)	(100,000)	(100,000)	(100,000)	
Management actions of utilising CBC (not cumulative)	90,000	-	40,000	10,000	-	-	140,000
Stress (after management action - only using CBC)	90,000	110,000	10,000	20,000	40,000	70,000	
Additional management actions to rebuild buffer (not cumulative)	-	-	-	30,000	20,000	10,000	60,000
Stress (after management action - incl. rebuilding HQLA buffer)	90,000	110,000	10,000	50,000	80,000	120,000	Total : GBP 200,000

Survival days, as set by the user within the 'liquidity stress assumptions' template (see Section 4.7) are presented in a separate table. The survival days both before and after management actions are presented, along with an analysis of the low points experienced during the stress. In the example shown below, a low point of £10 million was identified, on the 25th February 2023, which is on Day 150.

(Impact of management action on survival days)	Before	After
Survival days	-	366
Shortfall date	01 Oct 2022	> 01 Oct 2023
Lowest point in 1 year (amount)	(10,000)	
Lowest point date	27 Feb 2023	
Lowest point day	D150	

The example also shows that after taking the necessary management actions, the bank would have survival days of at least 366 days.

One of the key elements of this Summary section is the *'Daily cumulative stress mismatch analysis'* chart, as demonstrated below:



This chart provides the cumulative mismatch, based on the daily cashflow analysis (as set by the user – see section 4.3). The chart models the cumulative inflow/(outflows) before the use of any management actions (red line), after the use of sufficient management actions to both cover any liquidity shortfalls and rebuild the buffer (green line) and after full use of counterbalancing capacity along with the additional action taken to rebuild the buffer (purple line). The full use of the counterbalancing capacity along with any other action taken is solely used to demonstrate the bank has a significant buffer of marketable assets (both LCR and non-LCR qualifying) that could be realized in a short timeframe to provide sufficient liquid, way over and above the requirement.

7.3 NSFR

A brief table to show the NSFR before and after stress is displayed in the section. An example is shown below, where the NSFR increases as short-term funds are largely withdrawn and the remaining funds are mainly long-term (>1 year) deposits, which attract a higher RSF & ASF factor:

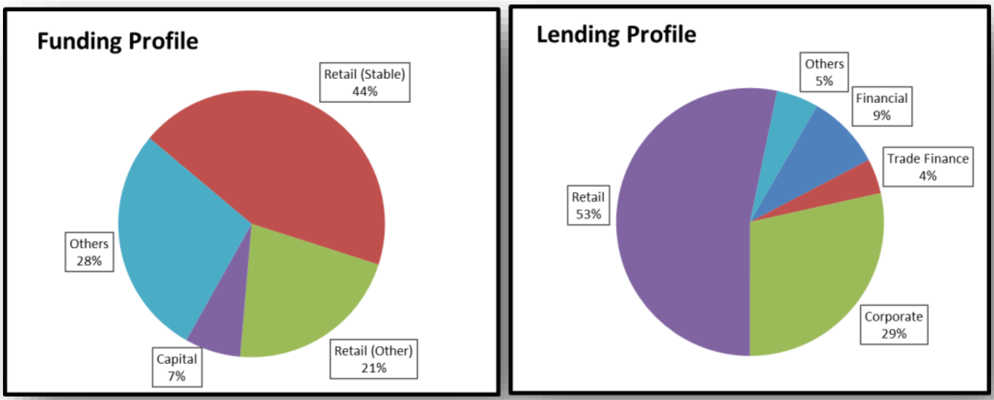
Net Stable Funding Ratio	Contractual Assessment	Stress Assessment
Available Stable Funding (ASF)	2,100,000	1,700,000
Required Stable Funding (RSF)	1,200,000	1,300,000
Net Stable Funding Ratio (NSFR)	175%	131%

7.4 Assets & Liabilities Profile

The summary report also includes the same Assets and Liabilities tables (provided in the 'liquidity results' report to show the before and after impact on the table, both after the impact of the stress and corresponding management actions. This provides a clear view to the reader of how certain portfolios are changing due to the stress event and action taken.

An example of the tables can be found in Section 5.3.

In addition to the tables, this section also illustrates the Funding and Lending profiles before stress in form of pie charts. This provides a visual image of the composition of the Bank's core portfolios. An example of the charts are provided below:



Katalysys contact details

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